Supporting information: Exact canonicalization of configured molecular stereographs

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A Detailed local algebra and information boundaries

A.1 Frame action and unique alignments

Lemma S1 (Frame action)**.** *Let u be a frame of length k, let $p, q \in S_k$, and let r be an injective relabeling of references. Then*

$$\mathrm{id} \cdot u = u, \qquad p \cdot (q \cdot u) = (q \circ p) \cdot u = (q \text{ then } p) \cdot u, \qquad r(p \cdot u) = p \cdot (ru).$$

Hence Eq. (4) is a right action of S_k on frames, the orbit $[u]_\Gamma = \{\gamma \cdot u : \gamma \in \Gamma\}$ is well defined for every subgroup $\Gamma \leq S_k$, and two witnesses are sequenced by p then q: $(p \text{ then } q) \cdot u = q \cdot (p \cdot u)$.

Proof. Coordinatewise, $(p \cdot (q \cdot u))_i = (q \cdot u)_{p(i)} = u_{q(p(i))} = ((q \circ p) \cdot u)_i$, which is the stated anti-homomorphism law and is exactly the convention $p \text{ then } q = p \circ q$ of the slot images. Thus the `then` API sequences tuple reorderings, although slot-map composition has the reverse order from composition of those reorderings. A relabeling acts on the values held by the slots and the group acts on the slots themselves, so $(r(p \cdot u))_i = r(u_{p(i)}) = (ru)_{p(i)} = (p \cdot (ru))_i$. Injectivity of $p \cdot u$ and of ru is inherited from u and r. □

Lemma S2 (Unique alignment)**.** *Let u and v be frames of length k with $\operatorname{im} u = \operatorname{im} v$. Then there is exactly one $\rho \in S_k$ with $v = \rho \cdot u$, namely $\rho = u^{-1} \circ v$. In particular every subgroup $\Gamma \leq S_k$ acts freely on frames, so all orbits have cardinality $|\Gamma|$.*

Proof. Both tuples are bijections onto the same image, so $\rho = u^{-1} \circ v$ is a permutation of slots and $(\rho \cdot u)_i = u_{\rho(i)} = v_i$. If $v = \rho' \cdot u$ as well, then $u_{\rho(i)} = u_{\rho'(i)}$ for every i, and injectivity of u forces $\rho = \rho'$. Freeness is the case $v = u$, which yields $\rho = \mathrm{id}$, and the orbit–stabilizer theorem then gives $|[u]_\Gamma| = |\Gamma|$. □

The tetrahedral case is the smallest instance of this local state model. Four distinct ligand references admit $4! = 24$ frames, on which S_4 acts freely and transitively; the subgroup A_4 has index two, so the frames split into two orbits of twelve. Even permutations are rigid rotations of one local configuration and odd permutations reach the mirror orbit.

Proposition S3 (Local parity criterion)**.** *Let u and v be tetrahedral frames with $\operatorname{im} u = \operatorname{im} v$, and let $\rho \in S_4$ be the unique alignment of Lemma S2, $v = \rho \cdot u$. Then*

$$[u]_{A_4} = [v]_{A_4} \quad \Longleftrightarrow \quad \operatorname{sgn}(\rho) = +1.$$

If $\operatorname{sgn}(\rho) = -1$, the two frames lie in the two distinct cosets of A_4 in S_4.

Proof. By Lemma S2 the alignment ρ exists and is unique, so $[u]_{A_4} = [v]_{A_4}$ holds precisely when $\rho \in A_4$. The sign homomorphism $\operatorname{sgn} : S_4 \to \{\pm 1\}$ has kernel A_4, whence $\rho \in A_4$ is equivalent to $\operatorname{sgn}(\rho) = +1$. Since $[S_4 : A_4] = 2$, every odd alignment lies in the unique remaining coset and therefore reaches the opposite orbit. □

A.2 Focal-configuration-neutral carrier perception

Perception produces typed *potential* elements, not configured descriptors. Let $\mathcal{F}$ be an optional set of supplied oriented frames at other loci. The focal frame is removed before its carrier is assessed, and all CIP properties, coordinates, canonical local descriptors, and atom indices are excluded. Algorithm S1 makes this boundary explicit.

Definition 1 (Potential-carrier predicate)**.** For a family policy P, identity profile χ, type τ, and candidate support s, let

$$\mathrm{Pot}_{\tau,P}(G_\chi, s; \mathcal{F}_{-s})$$

mean that s is a structurally valid support of type τ and passes the registered configuration-neutral feasibility test. The returned evidence record is

$$e_s = (s, \tau, A_s, q_s, r_s, \bot, \textsc{unassessed}),$$

where A_s is the action induced by the support stabilizer $\mathrm{Aut}_\chi(G, \mathcal{F}_{-s})_s$, q_s is the potential-carrier status, r_s is its reason, $\bot$ denotes absent configuration, and the final field records that configurational persistence has not been assessed.

The predicate is deliberately an over-approximation of experimentally persistent stereochemistry. It asserts neither a handedness nor a molecular chirality verdict. Its mathematical requirement is equivariance: if $f \in \mathrm{Iso}_\chi(G, H)$, then

$$\mathrm{Pot}_{\tau,P}(G_\chi, s; \mathcal{F}_{-s}) \Longleftrightarrow \mathrm{Pot}_{\tau,P}(H_\chi, f(s); f\mathcal{F}_{-s}), \qquad f_* A_s f_*^{-1} = A_{f(s)}. \tag{S1}$$

Algorithm S1 Focal-configuration-neutral stereo-carrier perception

Require: LLG G, identity profile χ, optional neighboring frames $\mathcal{F}$, family policy P
Ensure: Equivariant potential-element evidence with no focal configuration; serialization order is not identity

$G_0 \leftarrow \textsc{EraseDerivedAndConfiguredState}(G_\chi)$
$\mathcal{C} \leftarrow \textsc{EnumerateTypedSupports}(G_0, P)$
$\mathcal{E} \leftarrow [\,]$
for all $s \in \textsc{CanonicalSupportOrder}(\mathcal{C})$ **do**
  if $\neg\textsc{ValidateSupport}(G_0, s)$ **then**
    $\mathcal{E} \leftarrow \mathcal{E} \parallel (s, \textsc{rejected}, \textsc{invalid support})$
    continue
  end if
  $\mathcal{F}_{-s} \leftarrow \textsc{RemoveFocalFrame}(\mathcal{F}, s)$
  $G_s \leftarrow \textsc{AddNeighborFrameConstraints}(G_0, \mathcal{F}_{-s})$
  $A_s \leftarrow \textsc{InducedStabilizerAction}(\mathrm{Aut}(G_s)_s)$
  $(q, r) \leftarrow \textsc{ConfirmCarrier}(\mathrm{type}(s), G_s, s, A_s)$
  $\mathcal{E} \leftarrow \mathcal{E} \parallel (s, q, r, \bot, \textsc{unassessed})$
end for
return $\mathcal{E}$

Proposition S4 (Perception soundness)**.** *Assume every registered support enumerator, validator, and family confirmation test is invariant under χ-isomorphism transporting the supplied neighboring frame constraints. Algorithm S1 is then equivariant in the sense of Eq.* (S1)*. Its output contains no configuration that could depend on atom numbering, CIP priority, or the focal input orientation.*

Proof. Support enumeration and validation commute with f by assumption. Conjugation by f maps $\mathrm{Aut}_\chi(G_s)_s$ onto $\mathrm{Aut}_\chi(H_{f(s)})_{f(s)}$, hence maps the induced slot action and every invariant confirmation result onto its counterpart. Canonical support ordering changes only output order. The focal orientation and all derived naming data were erased, and the algorithm writes $\perp$ in every configuration field. □

ConfirmCarrier is geometry specific. A tetrahedral support is confirmed exactly when its center-stabilizer action contains no odd permutation of the four coordination slots. Planar-bond and axis candidates require non-exchangeable terminal reference pairs under the support stabilizer. Coordination carriers require a declared shape with valid arity. Path and plane carriers require their respective topology or authorized sidecar contract. Failed criteria retain the broad candidate with a structured reason such as `symmetry_related`. They do not imply a molecular achirality verdict. The output configuration is always $\perp$, and stability is always unassessed.

Terminal-reference equivalence must also remain separate from the orbit of a whole configured axis. A constitutional map can exchange both terminal pairs simultaneously, preserving axial orientation. Such a map supplies terminal equivalence witnesses without identifying a configured state with its mirror. The independent coupled-terminal regression checks this distinction. Exact configured-graph comparison retains the full joint frame action; a `symmetry_related` carrier reason alone cannot replace that comparison.

A.3 Authorized attachment and completeness boundaries

Perception and configuration attachment are separate typed operations. Let $\mathcal{D}$ be a finite collection of evidence records, let $\mathrm{src}(d)$ identify their provenance, and let $\mathrm{Auth} \subseteq \mathcal{D}$ be the records whose source is an explicit stereo encoding, a validated geometry, or a declared sidecar. Authorization is transported with the evidence record: for an isomorphism f, the target evidence collection and authorized subset are $f\mathcal{D}$ and $f\,\mathrm{Auth}$. Authorization and typing are not recomputed from atom indices. Every support validator invoked below must satisfy the same profile-relative invariance contract as perception.

Definition 2 (Authorized attachment)**.** Attachment is the partial map

$$\mathrm{att}\colon (\mathcal{X}, d) \longmapsto \big(G,\ \mathcal{R} \cup \{s_d\}\big),$$

defined exactly when the typing predicate $\mathrm{Typed}(d, \mathcal{X})$ holds, that is, when $d \in \mathrm{Auth}$; the support of d is a perceived or declared locus of the type τ that d declares; the ports of d coincide with P_s including owner scope for virtual references; the arity agrees with k_τ; the supplied state is a Γ_τ-orbit of an admissible frame in the sense of Definition 4; and $\mathcal{R}$ carries no conflicting configuration at that locus. If any conjunct fails, att is undefined and $\mathcal{X}$ is returned unchanged together with a structured refusal.

Lemma S5 (Attachment is equivariant and fail-closed)**.** *For every $f \in \mathrm{Iso}_\chi(G, H)$, $\mathrm{Typed}(d, \mathcal{X})$ holds if and only if $\mathrm{Typed}(fd, f\mathcal{X})$ holds, and in that case $\mathrm{att}(f\mathcal{X}, fd) = f\,\mathrm{att}(\mathcal{X}, d)$. Attaching the same configuration twice is idempotent, no undefined case modifies $\mathcal{R}$, and connectivity-only evidence, having $d \notin \mathrm{Auth}$, can never select a configuration orbit.*

Proof. Every conjunct of Typed is stated in terms of type, support, ports, arity, and orbit membership, all of which are transported by f; the orbit condition is preserved because f acts on references and commutes with the slot action by Lemma S1. The

image locus is therefore defined precisely when the source locus is, and the two constructions agree. Idempotence holds because $\mathcal{R} \cup \{s_d\}$ is a set union and the conflict conjunct rejects any second, different state at the same locus. The remaining claims are immediate from the definition, which modifies $\mathcal{R}$ only on the defined branch. □

Definition 3 (Support boundary and audit)**.** A *support boundary* B is an invariant rule specifying which configured loci must be represented for the requested claim. Important instances are B_{decl}, which contains exactly the source-declared configured loci, and B_{strict}, which contains all supported potential loci required by the active family policy. The third instance is B_{comp}, the population of supported configurations considered by topology completion. The audit

$$\mathrm{Audit}_B(G_\chi, \mathcal{E}, \mathcal{R}) \in \{\textsc{complete}, \textsc{incomplete}, \textsc{unsupported}\}$$

returns COMPLETE exactly when every locus required by B has one well-typed supported configuration and no required locus has unresolved or unsupported semantics.

Thus a binary verdict under B_{decl} is a claim about the represented source-declared stereograph. It becomes a claim about every supported stereogenic locus of the molecular input only under B_{strict}. Stating B prevents those two scientifically different conclusions from being conflated.

A.4 Finite-frame configuration orbits

For a finite geometry t, a local frame u is an ordered tuple of anchors and ligand references in the sense of Section 2.2.2, and its fixed configuration is the orbit

$$[u]_{\Gamma_t} = \{\gamma \cdot u : \gamma \in \Gamma_t\}.$$

The stored local state is the orbit, not a preferred tuple. An unspecified orientation uses a larger information group $\Gamma_t^{\uparrow}$, but this does not enlarge equality of a specified configuration.

Two groups act on the slots of a locus and must be distinguished. The larger one records which slot rearrangements are admissible at all; the smaller one records which of them leave the configuration fixed.

Definition 4 (Frame symmetry, admissible frames, geometry group)**.** Let t be a finite geometry of arity k_t. The *frame symmetry group* $\Sigma_t \leq S_{k_t}$ is the group of slot permutations compatible with the port structure of t. For an atom-supported geometry every ligand may occupy every coordination slot, so $\Sigma_t = S_{k_t}$. For the bond- and axis-supported families the slots consist of two terminal pairs together with their anchors, and Σ_t is the order-eight group generated by the two within-end exchanges and the end exchange that carries the anchors with it. The *admissible frames* of a locus with frame u are $\mathrm{Fr}_t(u) = \Sigma_t \cdot u$, and the geometry-preserving group is a subgroup $\Gamma_t \leq \Sigma_t$.

Proposition S6 (Configuration count)**.** *Let u be an admissible frame of a geometry t. The assignment*

$$\sigma\Gamma_t \longmapsto [\sigma \cdot u]_{\Gamma_t}$$

is a bijection from the left-coset space Σ_t/Γ_t onto the set of configurations of the locus. The number of configurations is therefore the index $[\Sigma_t : \Gamma_t] = |\Sigma_t|/|\Gamma_t|$, independently of the representative u.

Proof. By construction Σ_t acts transitively on $\mathrm{Fr}_t(u)$, and by Lemma S2 it acts freely, so $\sigma \mapsto \sigma \cdot u$ is a bijection $\Sigma_t \to \mathrm{Fr}_t(u)$. Lemma S1 gives

$$\Gamma_t \cdot (\sigma \cdot u) = \{(\sigma \circ \gamma) \cdot u : \gamma \in \Gamma_t\} = (\sigma\Gamma_t) \cdot u,$$

so this bijection carries left cosets of Γ_t exactly onto Γ_t-orbits. Counting cosets gives the index, and a different representative $u' = \sigma_0 \cdot u$ only translates the coset labels. □

Proposition S6 reproduces the state counts of Table 3 without case analysis: $24/12 = 2$ tetrahedral, $24/8 = 3$ square planar, $120/6 = 20$ trigonal bipyramidal, $720/24 = 30$ octahedral, and $8/4 = 2$ for the planar bond, the atrop axis, and the cumulene axis. It distinguishes a unique binary opposite from a mirror operation that requires additional registered geometric structure.

Corollary S7 (Binary opposite)**.** *If* $[\Sigma_t : \Gamma_t] = 2$*, then* $\Gamma_t \trianglelefteq \Sigma_t$*,* $\Sigma_t/\Gamma_t \cong \mathbb{Z}/2$*, and*

$$\mathrm{opp}([u]_{\Gamma_t}) = [\mu \cdot u]_{\Gamma_t}, \qquad \mu \in \Sigma_t \setminus \Gamma_t,$$

is independent of the choice of μ *and satisfies* $\mathrm{opp} \circ \mathrm{opp} = \mathrm{id}$*. If* $[\Sigma_t : \Gamma_t] > 2$*, the index alone selects no unique other state; an additional normalizer or improper geometric action can still define a mirror involution.*

Proof. A subgroup of index two is normal and its quotient is the two-element group. All elements of $\Sigma_t \setminus \Gamma_t$ form the single nontrivial coset, so $[\mu \cdot u]$ does not depend on μ, and applying the nontrivial coset twice returns the trivial one. When the index exceeds two, the quotient set has more than one other state, so the index-two uniqueness argument no longer applies. This does not exclude additional invariant structure: an element $\mu \in N_{\Sigma_t}(\Gamma_t)$ with $\mu^2 \in \Gamma_t$ induces a well-defined involution $[u] \mapsto [\mu \cdot u]$. The registered TBP and octahedral improper actions are examples despite their indices 20 and 30. □

For tetrahedral geometry Σ_t/Γ_t is the sign quotient S_4/A_4, so Corollary S7 specializes to Proposition S3. The same index-two argument applies to the planar bond and to both axis families. A configuration-changing permutation at a square-planar, trigonal-bipyramidal, or octahedral center is by contrast a general reconfiguration in the local relation API. Its separately registered geometric mirror may nevertheless supply an involution. Figure S1 gives one representative move per atom- or bond-supported geometry; full orbit enumerations are given in Appendix H.2.

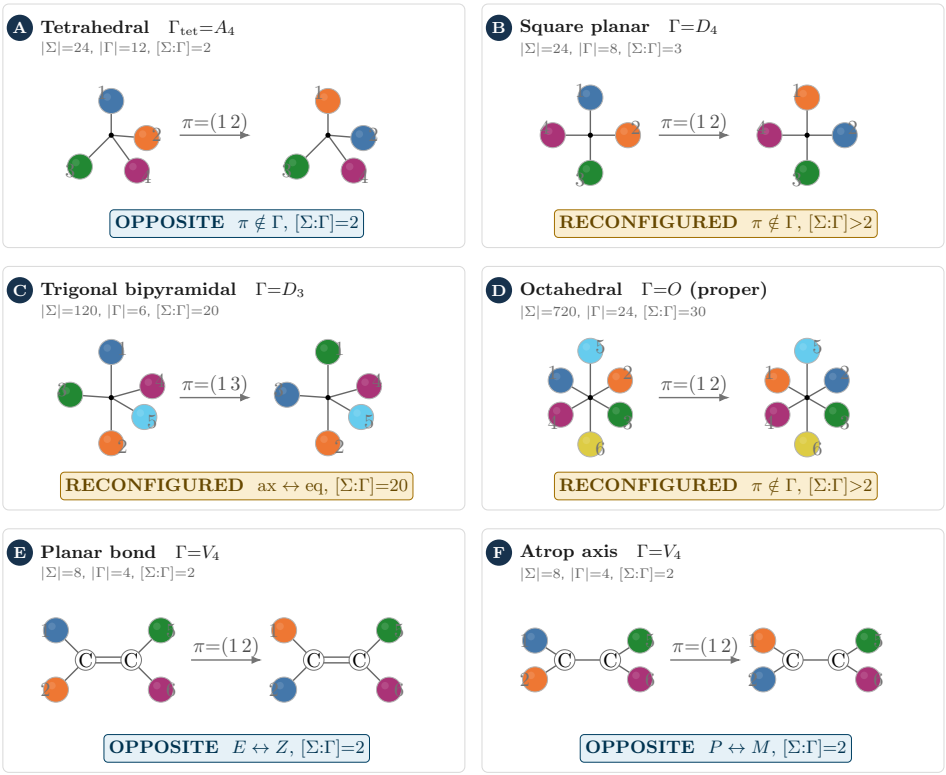


Fig. S1 One representative permutation π per atom- or bond-supported geometry. Balls are ligand references and the small digits are frame slots, so only the assignment of references to slots changes across each arrow. Each header lists the frame symmetry group Σ_τ, the geometry group Γ_τ, and their index, which by Proposition S6 is the number of configurations. Blue marks `OPPOSITE` and orange marks `RECONFIGURED` in the sense of Definition 5. By Corollary S7, a canonical binary opposite exists from the index alone for the index-two families (A, E, F); for (B), (C), and (D) the displayed move is a general reconfiguration. This does not exclude the separately registered geometric mirror on TBP or octahedral states.

For planar bonds, a whole-bond reversal and simultaneous exchange of both terminal pairs preserve configuration. A one-end exchange does not. Figure S2 keeps this second fundamental permutation example in the supplementary algebra.

A.5 Configuration relations and witnesses

Let $A = [u]_{\Gamma_t}$ and $B = [v]_{\Gamma_t}$ be configurations of the same geometry on a common reference set, and let π be the unique alignment $v = \pi \cdot u$ of Lemma S2. A concrete transport depends on which representatives were selected; its double coset does not.

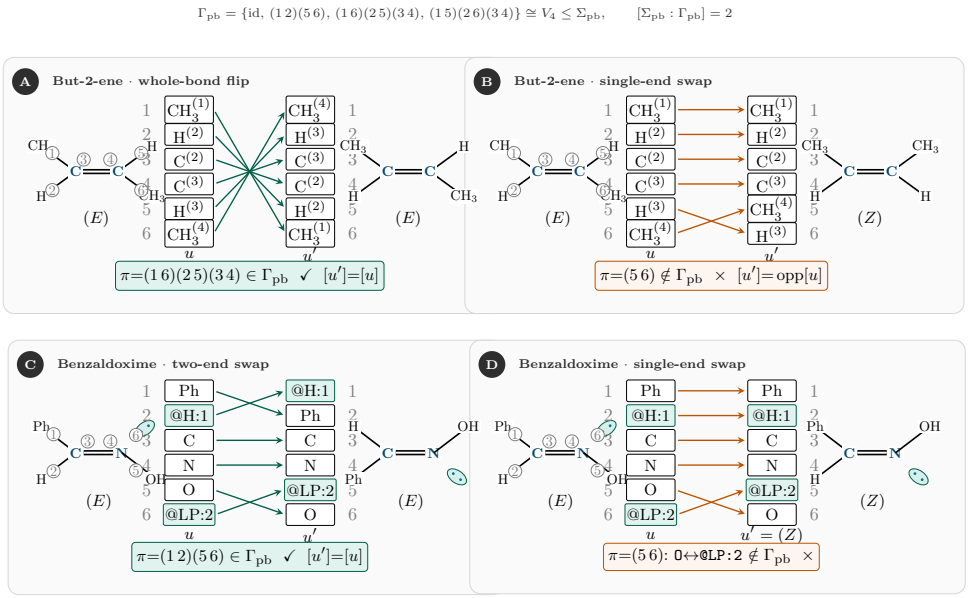


Fig. S2 Planar-bond frame algebra on but-2-ene (A, B) and benzaldoxime (C, D). Each panel aligns the ordered frame u of the left structure with the frame u' of the right structure slot by slot; slots 1, 2 and 5, 6 are the two terminal ligand pairs and slots 3, 4 are the bond anchors. (A) Reading the same E state from the opposite bond end is the whole-frame reversal $(1\,6)(2\,5)(3\,4) \in \Gamma_{pb}$, so the configuration is preserved. (B) The bare one-end swap $(5\,6)$ lies in $\Sigma_{pb} \setminus \Gamma_{pb}$ and therefore reaches the opposite orbit, $E \leftrightarrow Z$. (C) The simultaneous two-end exchange $(1\,2)(5\,6)$ is again in Γ_{pb}, here with the owner-scoped virtual references `@H:1` and `@LP:2` occupying ligand slots. (D) The same one-end swap is rejected when the moved slot carries the nitrogen lone pair. Green marks admitted alignments and vermilion rejected ones; the index $[\Sigma_{pb} : \Gamma_{pb}] = 2$ is what makes the rejected cases a binary opposite rather than an unrelated reconfiguration.

Lemma S8 (Representative independence of the relation)**.** *For $\alpha, \beta \in \Gamma_t$ put $u' = \alpha \cdot u \in A$ and $v' = \beta \cdot v \in B$. The unique alignment π' with $v' = \pi' \cdot u'$ is*

$$\pi' = \alpha^{-1} \circ \pi \circ \beta.$$

As α and β range over Γ_t, the alignments between representatives of A and of B are exactly the elements of the double coset $\Gamma_t \pi \Gamma_t$, which is therefore an invariant of the ordered pair (A, B).

Proof. Using Lemma S1 twice, $v' = \beta \cdot (\pi \cdot u) = (\pi \circ \beta) \cdot u$ and $u = \alpha^{-1} \cdot u'$, hence $v' = (\pi \circ \beta) \cdot (\alpha^{-1} \cdot u') = (\alpha^{-1} \circ \pi \circ \beta) \cdot u'$. Uniqueness of the alignment identifies

π' with this element. Since $\alpha \mapsto \alpha^{-1}$ is a bijection of Γ_t, the set of such π' is $\Gamma_t \pi \Gamma_t$; replacing u, v by other representatives replaces π by an element of the same double coset and leaves the set unchanged. □

Definition 5 (Relation kinds)**.** For same-geometry configurations on a common reference set, the stored relation kind is determined by $\Gamma_t \pi \Gamma_t$: `EQUIVALENT` when $\pi \in \Gamma_t$, equivalently $A = B$; `OPPOSITE` when $\pi \notin \Gamma_t$ and $[\Sigma_t : \Gamma_t] = 2$, so that $B = \mathrm{opp}(A)$ in the sense of Corollary S7; and `RECONFIGURED` when $\pi \notin \Gamma_t$ and $[\Sigma_t : \Gamma_t] > 2$. The kind is `UNRELATED` when the geometries or reference sets differ, so that no alignment exists, and `UNSPECIFIED` when either endpoint carries no configuration.

By Lemma S8 the kind and the double-coset identifier are representative independent, while the stored witness is a concrete permutation that replays one actual alignment. Witnesses compose by then as in Lemma S1: if p replays $A \to B$ and q replays $B \to C$, then p then q replays $A \to C$. Double cosets classify endpoint pairs and are not composed, because $\Gamma_t \pi \Gamma_t \cdot \Gamma_t \pi' \Gamma_t$ need not be a single double coset.

Stored specification is four-valued: specified, unknown, unspecified, and absent. Equality keeps these states distinct. Wildcard behavior belongs to an explicit query policy, never to stored molecular identity. Proofs of double-coset representative independence, witness composition, typed virtual reference ownership, and stereo-morphism policy laws are provided in Appendix H.2 through Appendix J.

A.6 From local parity to a global invariant

For one tetrahedral locus, Proposition S3 decides a candidate alignment by one group-membership test:

$$\rho \text{ even} \iff \rho \in A_4 \iff \rho \text{ preserves the local configuration.}$$

For a molecule, however, one atom bijection induces *all* local alignments simultaneously, and those choices are coupled whenever the constitution has automorphisms that exchange ligands or entire loci. The following lemma makes the induced family precise and records the law by which it composes.

Lemma S9 (Induced local alignments)**.** *Let $f \in \mathrm{Iso}_\chi(G, H)$ carry the typed carrier inventory of $\mathcal{X}$ onto that of $\mathcal{Y}$, let s be a finite-frame locus of $\mathcal{X}$ with frame u_s, and let $v_{f(s)}$ be a frame of the image locus. Applying f entrywise, $f(u_s)$ is an admissible frame of $f(s)$, so by Lemma S2 there is a unique*

$$\rho_s(f) \in \Sigma_{\tau_s} \qquad \text{with} \qquad f(u_s) = \rho_s(f) \cdot v_{f(s)}.$$

Then

1. *f carries the configuration at s to the configuration at $f(s)$ if and only if $\rho_s(f) \in \Gamma_{\tau_s}$, and this membership does not depend on the chosen representatives;*
2. *$\rho_s(\mathrm{id}) = \mathrm{id}$ and, for a second isomorphism g,*

$$\rho_s(g \circ f) = \rho_{f(s)}(g) \text{ then } \rho_s(f), \tag{S2}$$

so $\rho_{f(s)}(f^{-1}) = \rho_s(f)^{-1}$.

Proof. Entrywise application of a bijection preserves injectivity and the port structure, so $f(u_s)$ is admissible and the alignment exists uniquely. For (i), replacing u_s by $\alpha \cdot u_s$ and $v_{f(s)}$ by $\beta \cdot v_{f(s)}$ with $\alpha, \beta \in \Gamma_{\tau_s}$ replaces $\rho_s(f)$ by an element of $\Gamma_{\tau_s}\rho_s(f)\Gamma_{\tau_s}$ by Lemma S8, and that double coset meets Γ_{τ_s} if and only if it equals Γ_{τ_s}. For (ii), relabeling commutes with the slot action by Lemma S1, hence

$$(g \circ f)(u_s) = g\big(\rho_s(f) \cdot v_{f(s)}\big) = \rho_s(f) \cdot g(v_{f(s)}) = \rho_s(f) \cdot \big(\rho_{f(s)}(g) \cdot w_{g(f(s))}\big)\,,$$

and the action law $p \cdot (q \cdot w) = (q \circ p) \cdot w$ identifies the last expression with $\big(\rho_{f(s)}(g) \circ \rho_s(f)\big) \cdot w_{g(f(s))}$, which is Eq. (S2). Taking $g = f^{-1}$ gives the inverse law. □

Lemma S9 reduces the global question to a *simultaneous* membership condition: a constitutional isomorphism is stereochemically admissible exactly when $\rho_s(f) \in \Gamma_{\tau_s}$ holds at every locus at once. Because Eq. (S2) is a cocycle law and each Γ_{τ_s} is a subgroup, this condition is closed under composition and inversion, which is what makes the relation of Definition 1 an equivalence. The orbit gadget of Section 2.6.4 converts each membership condition into an ordinary graph-automorphism constraint; joining all gadgets to the same constitutional graph forces one global atom map to satisfy every constraint at once, and exact canonical labeling then quotients over all such maps instead of selecting local representatives in advance.

A.7 End-to-end decision from an LLG

Algorithm S2 makes the full contract explicit. Potential-carrier discovery precedes attachment; only typed authorized evidence may attach configuration; and no binary result is emitted until the declared boundary is complete. The exact mirror comparison in Algorithm 2 is then the final decision kernel.

Theorem S10 (Soundness and completeness of the binary verdict)**.** *Fix an identity profile χ, family policy P, and invariant boundary B. Assume that normalization under χ commutes with material relabeling and transports every retained carrier and authorized evidence record consistently. Assume also that (i) perception satisfies Proposition S4; (ii) attachment accepts only well-typed authorized configuration evidence and commutes with χ-isomorphisms; (iii) the boundary audit reports* COMPLETE *exactly for complete supported registries over B; (iv) the mirror action satisfies Eq. (8); and (v) the expansion and canonizer satisfy Theorems 4 and 5. Then Algorithm S2 is numbering invariant. On complete supported inputs for which both exact canonical*

Algorithm S2 Fail-closed molecular chirality from Lewis-labeled input

Require: LLG G; evidence $\mathcal{D}$; profile χ; family policy P; boundary B
Ensure: ACHIRAL, CHIRAL, INCOMPLETE, or UNSUPPORTED, with evidence

$G_\chi \leftarrow \textsc{NormalizeLLG}(G, \chi)$
if $\textsc{IsRefusal}(G_\chi)$ **then**
  return $\textsc{Unsupported}(\text{INVALID CONSTITUTION})$
end if
$\mathcal{E} \leftarrow \textsc{PerceivePotentialCarriers}(G_\chi, P)$
$\mathcal{R} \leftarrow [\,]$
for all $d \in \textsc{CanonicalEvidenceOrder}(\mathcal{D})$ **do**
  if $\neg\textsc{AuthorizedSource}(\mathrm{src}(d))$ **then**
    return $\textsc{Unsupported}(d, \text{UNAUTHORIZED CONFIGURATION SOURCE})$
  end if
  $r \leftarrow \textsc{ValidateAndAttach}(G_\chi, \mathcal{E}, d)$
  if $\textsc{IsRefusal}(r)$ **then**
    return r
  end if
  $\mathcal{R} \leftarrow \mathcal{R} \,\|\, r$
end for
$q \leftarrow \textsc{AuditBoundary}(B, G_\chi, \mathcal{E}, \mathcal{R})$
if $q \neq \text{COMPLETE}$ **then**
  return q with audit evidence
end if
$\mathcal{X}_B \leftarrow (G_\chi, \mathcal{R}, B)$
return $\textsc{ExactMirrorChirality}(\mathcal{X}_B)$

searches finish, its binary result obeys:

$$\text{ACHIRAL} \iff \mathcal{X}_B \cong_{\mathrm{st}} \mu_B \mathcal{X}_B,$$

$$\text{CHIRAL} \iff \mathcal{X}_B \ncong_{\mathrm{st}} \mu_B \mathcal{X}_B.$$

Every missing, ill-typed, unsupported, unauthorized, or search-bounded case returns a refusal rather than either binary value.

Proof. Equivariance of normalization, perception, attachment, and the boundary audit makes the constructed $\mathcal{X}_B$ independent of atom numbering. On a complete registry, mirror naturality makes $\mu_B\mathcal{X}_B$ another well-defined configured stereograph over the same constitutional profile. Theorem 5 states that the two full certificates are equal

exactly when these objects are stereo-isomorphic, which is the definition of achirality. Certificate inequality therefore proves chirality because the canonizer is complete. All earlier failed preconditions and any interrupted exact search are explicit non-binary exits. □

Remark 1 (Scope of the theorem). The theorem is relative to (χ, P, B). It proves chirality of the represented configured stereograph, not experimental configurational persistence, conformational population, or completeness of chemistry outside the registered family policy. Under B_{decl}, undeclared potential loci are intentionally outside the claim; under B_{strict}, they must be configured or the algorithm refuses.

A.8 Incomplete input and topology completion

A binary mirror verdict requires a complete, supported configured representation. A perceived carrier with no supplied orientation returns `INCOMPLETE`; unsupported enhanced-stereo population semantics return `UNSUPPORTED`. Connectivity alone never acquires a handedness.

A separate topology-completion assessor addresses a different question: across supported assignments of an underspecified constitution, is every completion achiral, every completion chiral, or is the outcome configuration-dependent? Its four outcomes are necessarily achiral, necessarily chiral, configuration dependent, and unsupported/incomplete. Let

$$\mathrm{Comp}_B(G_\chi, \mathcal{E}, \mathcal{D})/\cong_{\mathrm{st}}$$

be the finite set of symmetry-unique, supported configured completions consistent with the supplied evidence. Algorithm S3 classifies that population. Its one-sided outcomes require exhaustive enumeration or an exact proof that excludes the opposite class; reaching an enumeration cap is not such a proof.

Algorithm S3 Four-state classification of an underspecified constitution

Require: $G, \mathcal{D}, \chi, P, B_{\text{comp}}$; exact completion policy Q
Ensure: One four-state result with proof evidence
$(\mathcal{C}, \eta) \leftarrow$ SymmetryUniqueCompletions$(G, \mathcal{D}, \chi, P, B_{\text{comp}}, Q)$
if η.unsupported **then**
  return unsupported/incomplete with η
end if
$h_A \leftarrow \bot$; $h_C \leftarrow \bot$
for all $\mathcal{X} \in \mathcal{C}$ **do**
  $r \leftarrow$ ExactMirrorChirality$(\mathcal{X})$
  if IsRefusal(r) **then**
    return unsupported/incomplete with r
  else if $r =$ achiral **then**
    $h_A \leftarrow r$.witness
  else
    $h_C \leftarrow (r.C, r.\overline{C})$
  end if
  if $h_A \neq \bot \wedge h_C \neq \bot$ **then**
    return configuration-dependent with (h_A, h_C)
  end if
end for
if $\neg\eta$.exhausted $\wedge \neg$ExactOneSidedProof(η, h_A, h_C) **then**
  return unsupported/incomplete with η
end if
if $h_A \neq \bot$ **then**
  return necessarily-achiral with (h_A, η)
else if $h_C \neq \bot$ **then**
  return necessarily-chiral with (h_C, η)
else
  return unsupported/incomplete with empty population
end if

Proposition S11 (Soundness of four-state completion)**.** *For a nonempty supported completion population, assume that all required validations and mirror comparisons complete. If completion generation is exhaustive modulo stereo-isomorphism, or its one-sided shortcut supplies the required exact existence/non-existence proof, then Algorithm S3 returns* necessarily-achiral *exactly when every completion is achiral,* necessarily-chiral *exactly when every completion is chiral, and* configuration-dependent *exactly when both kinds exist. Two valid opposite-class witnesses suffice*

for the mixed result even without exhaustion. An empty population, unresolved validation, interrupted comparison, or uncertified one-sided population returns a refusal.

Proof. Theorem S10 decides each complete representative exactly, and chirality is constant on stereo-isomorphism classes. Exhaustion therefore quantifies over the entire completion population. Witnesses of both classes prove the mixed result immediately; a one-sided result is emitted only after the opposite class has been excluded. Every remaining state takes the explicit refusal branch. □

The current assessor uses provisional topology probes to reach globally stereogenic ACS cases not localized by the configured registry. Its result is reported separately from the exact source-declared certificate because the two procedures do not have the same representation boundary.

We additionally encode those probes as one coupled `GlobalStereoSupport` certificate. Topology may thereby prove a necessarily-chiral molecular outcome, while the associated `FrameworkStereo` orientation remains unspecified. Only explicit authorized sidecar or future validated coordinate evidence may select the positive or negative configured framework. This exact-plus-global protocol is versioned separately from both the source-declared audit and the ACS-specialized classifier.

B Additional mathematical and historical details

B.1 Lewis electron resource bookkeeping

Writing $o_e = s_e + p_e$, the valence-electron degree and formal charge are

$$n_v = 2\ell_v + r_v, \qquad d_v = n_v + \sum_{e \sim v} o_e, \qquad q_v = z_v - d_v, \tag{S3}$$

with z_v the element-specific valence-electron count. Equivalently, the electron loci

$$\mathrm{Loc}(G) = \{\mathrm{lp}(v), \rho_v : v \in V\} \cup \{\sigma_{uv}, \pi_{uv} : \{u, v\} \in \tbinom{V}{2}\}$$

have occupancies $c_G(\mathrm{lp}(v)) = \ell_v$, $c_G(\rho_v) = r_v$, $c_G(\sigma_{uv}) = s_{uv}$, and $c_G(\pi_{uv}) = p_{uv}$. Separating σ and π resources is essential, for example, to distinguish a cumulated double-bond path from a single-bond chain.

B.2 Raw formal and global assignment quotients

The benchmark terminology follows two different equivalence relations and is therefore not interchangeable. Fix a colored constitution G_χ, a finite typed carrier set S, and for every $s \in S$ a set $\mathcal{R}_s$ of admissible ordered raw frames. The registered preserving group Γ_{τ_s} acts on $\mathcal{R}_s$. Define

$$\mathcal{F}_s = \mathcal{R}_s/\Gamma_{\tau_s}, \qquad \mathcal{R} = \prod_{s\in S} \mathcal{R}_s, \qquad \mathcal{F} = \prod_{s\in S} \mathcal{F}_s,$$

and let $q = \prod_s q_s : \mathcal{R} \to \mathcal{F}$ be the componentwise local quotient. Thus an element of $\mathcal{R}$ chooses one ordered frame at each carrier, whereas an element of $\mathcal{F}$ chooses one formal local configuration orbit at each carrier. By Proposition S6 each factor has $|\mathcal{F}_s| = [\Sigma_{\tau_s} : \Gamma_{\tau_s}]$ elements and each fiber of q_s has exactly $|\Gamma_{\tau_s}|$ elements, so $|\mathcal{R}| = |\mathcal{F}| \cdot \prod_{s\in S} |\Gamma_{\tau_s}|$.

Let $A = \mathrm{Aut}_\chi(G, S, \tau)$ be the automorphism group of the colored constitution that preserves the typed carrier inventory. Every $\alpha \in A$ transports the ports of s to those of $\alpha(s)$ and therefore induces a map $T_{\alpha,s}$ on local states.

Lemma S12 (Transport is a group action)**.** *For $\alpha \in A$ and $s \in S$ put $T_{\alpha,s}\big([u]_{\Gamma_{\tau_s}}\big) = [\alpha(u)]_{\Gamma_{\tau_{\alpha(s)}}}$. Then $T_{\alpha,s}$ is well defined, $T_{\mathrm{id},s} = \mathrm{id}$, and $T_{\alpha\circ\beta,s} = T_{\alpha,\beta(s)} \circ T_{\beta,s}$.*

Consequently

$$(\alpha \cdot a)_{\alpha(s)} = T_{\alpha,s}(a_s)$$

defines an action of A on $\mathcal{F}$, and the whole-molecule configuration classes for this fixed substrate are the orbit set $\mathcal{C} = \mathcal{F}/A$.

Proof. Well-definedness is the commutation law of Lemma S1: for $\gamma \in \Gamma_{\tau_s}$ we have $\alpha(\gamma \cdot u) = \gamma \cdot \alpha(u)$, so an entire orbit maps into a single orbit, and α preserves the type $\tau_s = \tau_{\alpha(s)}$ because it preserves the typed carrier inventory. Functoriality and the identity law hold because α acts entrywise on references. These two laws are exactly the conditions for the displayed formula to be a group action on the product $\mathcal{F}$. □

Because A and $\mathcal{F}$ are finite, the number of global classes is given by the orbit-counting (Cauchy–Frobenius) lemma,

$$|\mathcal{C}| = \frac{1}{|A|} \sum_{\alpha \in A} |\mathrm{Fix}_{\mathcal{F}}(\alpha)|.$$

The first quotient removes only redundant descriptions of each local state; the second resolves exchange of ligands and of complete carriers by one global constitutional automorphism. Unlike the local quotient, the second is in general not free, so $|\mathcal{C}|$ is not $|\mathcal{F}|/|A|$; fixed points are exactly the internally compensated assignments, of which *meso* forms are the familiar case. In particular $|\mathcal{F}|$ is not generally the number of stereoisomers, and raw multiplicity is not a new chemical degree of freedom.

B.3 Exact raw enumeration factorization

Corollary S13 (Exact factorization of raw enumeration)**.** *For assignments $a, b \in \mathcal{F}$ on the fixed typed substrate above,*

$$C_{\mathrm{SG}}(\mathcal{X}(a)) = C_{\mathrm{SG}}(\mathcal{X}(b)) \quad \Longleftrightarrow \quad b = \alpha \cdot a \text{ for some } \alpha \in A.$$

Consequently, for raw encodings $r, r' \in \mathcal{R}$,

$$C_{\mathrm{SG}}(\mathcal{X}(q(r))) = C_{\mathrm{SG}}(\mathcal{X}(q(r'))) \quad \Longleftrightarrow \quad [q(r)]_A = [q(r')]_A.$$

Thus the raw route $\mathcal{R} \xrightarrow{q} \mathcal{F} \to \mathcal{C}$ and direct formal enumeration compute the same global partition.

Proof. An isomorphism between two configured stereographs over the same substrate restricts to an element of A, and Definition 1 requires its induced port transports to carry every component of a to the corresponding component of b. Conversely, any such element of A extends through every complete local-orbit gadget and is a stereo-isomorphism. The first equivalence now follows from Theorem 5; substituting $a = q(r)$ and $b = q(r')$ proves the second. □

Corollary S13 makes formal enumeration exact rather than heuristic compression. Raw enumeration remains valuable as an independent executable check that construction, transport, and serialization implement q consistently; it is not required to repeat every member of a local orbit once the quotient law and its implementation gates hold.

B.4 Automorphism projection

Proposition S14 (Automorphism projection)**.** *Under the admissibility and uniqueness conditions of Theorem 4, restriction to atom vertices is a group isomorphism*

$$\mathrm{proj}_G : \mathrm{Aut}(\Phi_\chi(\mathcal{X})) \xrightarrow{\sim} \mathrm{Aut}_{\mathrm{st}}(\mathcal{X}).$$

Proof. Restriction is a group homomorphism because composition of auxiliary automorphisms restricts to composition of their atom parts. It lands in $\mathrm{Aut}_{\mathrm{st}}(\mathcal{X})$ and is surjective by the two directions of Theorem 4 applied with $\mathcal{Y} = \mathcal{X}$. It is injective because an auxiliary automorphism fixing every atom vertex fixes every other vertex:

an incidence vertex is determined by its atom endpoints and type color. A locus is determined by its full typed carrier key, recovered from the support incidences and ordered support relations, including any pilot and declared coupling identity. The atom set alone need not distinguish two paths through the same atoms. The unique-key condition therefore excludes exchanging two identical carrier gadgets. A port vertex is determined by its owner and referent, a tuple vertex by the sequence of ports it reaches through position-colored slots, and a slot vertex by the pair it joins together with its position color. Private virtual resources are fixed by their typed owner and attached slot. A bijective homomorphism of groups is an isomorphism. □

Consequently the stereo-aware atom orbits reported by the engine are the orbits of $\mathrm{Aut}_{\mathrm{st}}(\mathcal{X})$ itself, not an approximation of them, and $|\mathrm{Aut}_{\mathrm{st}}(\mathcal{X})|$ is available from the same search that produces the certificate.

B.5 Historical representation and relation gates

Historical raw products and secondary relation gates are retained here as development evidence. Their archived invariance and multiplicity checks supply development evidence; they do not independently establish the faithfulness of the encoding. Historical raw A collapses 2,048 encodings onto 48 formal tuples and 38 global classes; raw B collapses 992,256 encodings onto 192 formal tuples and 95 classes. The complete raw C/VS146 slice collapses 2,359,296 encodings onto its 64 formal tuples and 64 classes. Each raw result matches the corresponding formal partition exactly, as predicted by Corollary S13. The multi-element experiment shows why this is a global rather than merely local test: 32 assignments reduce to 31 classes because one designed symmetry identifies two assignments, and the complete mirror relation matrix is obtained from the same certificates.

Table S2 gives the complete mixed-family accounting. Formal enumeration covers every A/B/C case. Raw A and B are complete, while raw C is complete only for the

Table S1 Canonicalization and relation results.

Gate	Result and interpretation
Repository regression	3881 passed, 6 skipped, and 179 subtests on the development snapshot.
Local quotient	940 representations yield 67/67 expected classes across ten configured families.
Exact class separation	67/67 classes are distinct within family, with zero collision or timeout.
Configuration-free local	66468/66468 representations and 3476/3476 carriers pass over 1218 rows; direct transport partitions and every positive map are independently checked.
Formal A/B/C	2160/2160 assignments pass over 30 cases; 38/95/752 global classes and complete reference partition agreement.
Historical raw quotient	A: 2,048/2,048; B: 992,256/992,256; C/VS146: 2,359,296/2,359,296 encodings, each with archived production-code/formal quotient agreement; independent negative-separation truth was not part of those historical gates.
Relabeling	201 per-class and 528 representative checks give zero failure.
Mirror closure	67/67 classes close under the geometry-specific mirror; seven are fixed.
Public permutations	888/888 exhaustive atom permutations of four ACS molecules are invariant.
CIP-cache removal	4/4 certificates are invariant, confirming that nomenclature is excluded.
Multi-element composition	32 assignments yield 31 classes, 13 enantiomer pairs, 64 diastereomer pairs, and five mirror-fixed classes.
Public multi-element audit	14/14 renumberings and 7/7 mirror labels pass on seven ACS records with 18 elements.
Pairwise matrix	9/9 identity, enantiomer, diastereomer, constitution, and incomplete cases are correct.

declared VS146 slice; the remaining eight C raw products include a largest Cartesian product of 782,757,789,696 and are intentionally not claimed as executed. Complete local orbits are still materialized before an automorphism-pruned individualization–refinement search. The archived wall times document the executed artifacts but are not normalized performance comparisons across budgets, cases, or machines.

The current formal and historical raw runs are complete for their declared scopes, with zero reported case failure or timeout. Current formal reports require independently checked complete assignment partitions. Historical raw reports check representation invariance and production-code quotient agreement. Their execution

Table S2 Mixed-family raw–formal–global quotient results. "Formal tuples" is the number covered invariantly by the run; "classes" counts distinct whole-stereograph certificates. C/VS146 is a scoped raw stress result, not the full C raw product. Elapsed time is archival wall time and is reported only for provenance.

Budget	Mode	Scope	Cases	Enumerated	Formal tuples	Classes
A	Formal	current complete A	10	48	48	38
A	Raw	historical complete A	10	2,048	48	38
B	Formal	current complete B	11	192	192	95
B	Raw	historical complete B	11	992,256	192	95
C	Formal	current complete C	9	1920	1920	752
C	Raw	historical VS146 only	1	2,359,296	64	64
Historical raw evidence			22	3,353,600	–	–

times are not a controlled comparison and are omitted here. The formal totals are 30/30 cases and 2,160/2,160 assignments. The raw totals are 22/22 scoped cases and 3,353,600/3,353,600 assignments. These data empirically check the local quotient on more than three million redundant encodings; the proof that formal enumeration preserves global classes is independent of that sample size.

C Supplementary chemical carrier gallery

Figure S3 provides representative chemical cases for the broader descriptor-class set. The examples identify the kind of carrier. They do not by themselves establish a rotational barrier, configurational lifetime, or implemented perception capability.

D Supplementary cost and resource measurements

E Scope and claim boundary of the technical appendices

These appendices consolidate the technical contracts for the molecular-graph layer of `SynKit`'s configuration-neutral stereochemical identity model: carrier perception and constitutional evidence, typed supports, frames, groups, orbits, relation classes, witnesses, stereo-preserving morphisms, graph identity, derived-label projection, and

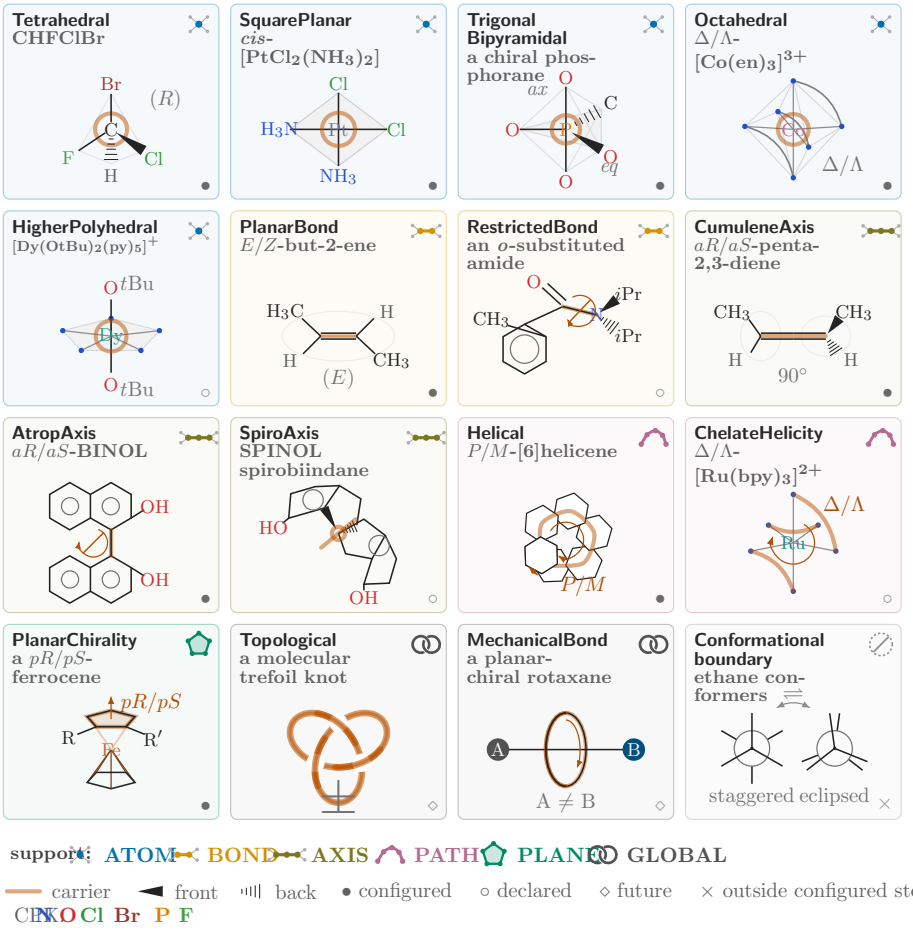


Fig. S3 Representative chemical instances of the descriptor classes. Card names abbreviate the `SynKit` classes ⟨**Name**⟩**Stereo**; the card tint and the glyph in its upper right give the support family, and vermilion marks the carrier of the stereo element. The corner mark states status: ● configured, ○ declared, ◇ future, and × for an entry that is deliberately outside configured stereo. The examples identify the kind of carrier only. They do not by themselves establish a rotational barrier, a configurational lifetime, or an implemented perception capability. Ordinary rapidly interconverting *s*-cis/*s*-trans conformers remain outside configured stereo, and a restricted-bond descriptor requires independent barrier or persistence evidence.

whole-molecule chirality. Reaction-layer operations, including rule guards and effects, reference-aligned stereo change, reversal, imaginary transition states and mechanism replay, are outside the present validation scope.

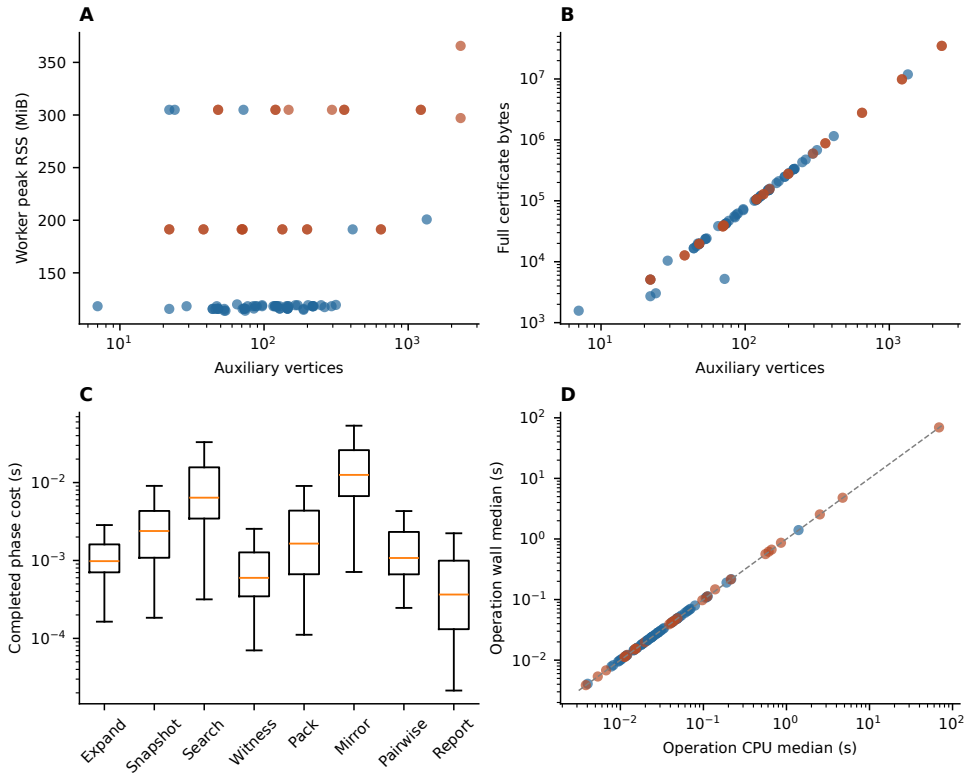


Fig. S4 Completed generator-mode resource measurements. (A) Median native worker peak RSS, including interpreter and imported libraries. (B) Full certificate size. (C) Per-trial phase distributions; boxes give quartiles, whiskers use the standard 1.5-interquartile-range rule and outliers are omitted only from this panel. Pairwise timing covers its declared six-family subset; report timing serializes the worker's measurement payload. (D) Operation CPU versus wall medians, excluding interpreter startup. Blue denotes ordinary cases and orange stress cases. Censored trials are absent from these panels and retained in Table 6.

The novelty boundary is explicit. Ordered local frames modulo finite geometry groups are prior MØD work, while permutation-invariant local descriptors and whole-graph stereoisomer comparison are also provided by `StereoMolGraph` [1, 2]. This work does not claim either premise as new. Its claim is the evidence-bearing contract joining configuration-neutral chemical-neighbourhood frames and carrier-stabilizer tie certificates to typed configured states, double-coset relations, composable transport witnesses, exact global equivalence, and derived labels that cannot feed back into identity. The complete configured object is additionally encoded as a coloured auxiliary

stereograph whose exact canonical label is the exact configured-stereograph identity certificate.

The SACA comparison uses its accessible primary abstract and the authors' inspected experiment source, rather than attributing guarantees from an unavailable full text [3]. Its resonance-invariant circular fingerprints are credited separately from the full-certificate identity contract studied here.

The method is compatible with the ordered-local-frame and finite-group model used by MØD; it is not claimed to be a complete monomorphic Double-Pushout implementation, a three-dimensional perception engine, or a predictor of stereochemical outcome.

Eight orbit shapes are registered—tetrahedral, square-planar, trigonal-bipyramidal, octahedral, planar-bond, atrop-axis, cumulene-axis, and extended-cis/trans—together with configured helical-path and planar-chirality descriptors. Global support is implemented by the coupled-frame `FrameworkStereo` descriptor and an orientation-unspecified analysis certificate. Section I.3 states exactly which consumers accept which classes.

F Reproducibility and environment manifest

Table S3 Research artifacts and their provenance.

Artifact	Recorded revision	Retained evidence
`SynKit` source	Package metadata `1.6.2`; clean experiment commit `e2526b9`	Exact source and input hashes; complete raw execution archive
Core fixtures	`Test/Graph/Stereo`, `Test/Graph/Morphism`, `Test/Chem/Molecule`	Frozen cases and source checksums in the research archive
Recorded environment	Linux, Python 3.12.12; 46-package base and separate comparator overlay	Wheel-hash locks and original conda bootstrap specification
Optional comparator	Current `StereoMolGraph 2b99f62`; historical `2189f61` remains separate	Pinned comparison reports and MIT notice; no runtime dependency
Benchmark registry	`Experiment/Stereo/Data/manifest.json`, schema `synkit.stereo-benchmark-registry/1`	Dataset revisions, input hashes, attribution and separate source licences

Descriptor equality, finite group actions, support projection, morphism construction, CIP ranking, and label projection are deterministic and use no random seed. Stereoisomer enumeration for the chirality benchmark is capped and the cap is reported with the result.

G Typed supports

Support answers where a stereo element lives. It deliberately does not encode configuration, CIP labels, provenance, or configurational stability.

Table S4 Support families, their data, and their validation invariants.

Family	Data	Rejects
Atom	one centre	non-integer centre
Bond	two endpoints	equal endpoints
Axis	ordered path of ≥ 2 atoms; two ordered two-reference terminal frames	repeated path atoms; wrong frame count or width; duplicate references inside one terminal frame
Path	ordered atoms; open/cyclic flag	fewer than two atoms; repeated atoms; cyclic with fewer than three atoms
Plane	≥ 3 oriented atoms	fewer than three atoms; repeated atoms
Global	component atom set	empty or non-integer atom set

Every support exposes a dependency set—the atoms whose renumbering affects it—and a relabelling action. Axis-support relabelling covers the path, the material references in both terminal frames, and the owners of virtual references, so that `@H:<owner>` and `@LP:<owner>` follow their atom.

H Permutation-orbit data model

H.1 Frames, groups, and canonical identity

A permutation p acts by

$$p(N)_i = N_{p(i)},$$

and `p.then(q)` has slot image $p \circ q$, so `p.then(q).apply(N)` equals `q.apply(p.apply(N))`. Atom-supported frames place the locus first and then their ligand references. Bond- and axis-supported frames use $(l_1, l_2, l, r, r_1, r_2)$. Material references are atom maps; virtual ligands have the owner-scoped forms `@H:<owner>` and `@LP:<owner>`.

For a registered shape τ with geometry group Γ_τ and frame N, a fixed configuration is the orbit $[N]_{\Gamma_\tau}$, and the number of configurations is the index $[\Sigma_\tau : \Gamma_\tau]$ of Proposition S6. The canonical representative is the minimum deterministically encoded orbit member. Two configurations are equal exactly when shape, specification level, and canonical frame agree.

Table S5 Registered fixed and unspecified group actions.

Shape	Support	Fixed group	Unspecified information group
Tetrahedral	atom	A_4, order 12	S_4, order 24
Square planar	atom	D_4, order 8	S_4, order 24
Trigonal bipyramidal	atom	D_3, order 6	S_5, order 120
Octahedral	atom	proper O, order 24	S_6, order 720
Planar bond	bond	V_4, order 4	bond-frame group, order 8
Atrop axis	axis	V_4, order 4	bond-frame group, order 8
Cumulene axis	axis	V_4, order 4	bond-frame group, order 8

Planar-bond, atrop-axis, and cumulene-axis fixed actions are distinct even though all three are abstractly V_4: they differ in their frame-reversal action and in the support that may carry them. Stored unspecified orientation uses the larger group and is distinct from descriptor absence and query wildcard behavior. The legacy `atoms`, `parity`, and `provenance` fields remain compatible encodings; parity is decoded to a concrete frame and is not mathematical identity. Existing `canonical_form()` and serialized descriptor payloads remain the Beta-2 compatibility boundary, and `AtropBondStereo` remains a wire-compatible alias of `AtropAxisStereo`.

H.2 Permutation orbits of the six atom- and bond-supported geometries

The tetrahedral orbit structure is drawn exhaustively in the main text: all $4! = 24$ orderings of four distinguishable ligands split into two 12-element orbits (the even rotations A_4 and their odd mirror images), so a stored descriptor is the orbit, in practice the parity. Figures S5–S9 apply the same construction to the remaining five geometries, using the same colour-for-ligand convention; each cell is one frame ordering, labelled by the permutation that produces it from the reference in cycle notation over the numbered slots. All cells, groupings, and labels were generated directly from the registered groups of Table S5. Where the full ordering set is too large to draw (trigonal-bipyramidal, $5! = 120$; octahedral, $6! = 720$), one configuration's orbit—the frame maps belonging to Γ_τ—is shown and the total configuration count $[\Sigma_\tau : \Gamma_\tau]$ is stated. The cumulene-axis frame is not drawn: its group and orbit structure coincide with the atrop frame of Figure S9, and it differs in support and admissibility rather than in group action.

H.3 Relations and witnesses

If a concrete permutation π transports one fixed frame to another, the representative-independent relation class is the double coset $\Gamma_\tau \pi \Gamma_\tau$ of Lemma S8. `StereoRelation` stores a stable minimum class identifier and a concrete `PermutationWitness`. Witnesses compose; double-coset identifiers do not. The public relation kinds are `EQUIVALENT`, `OPPOSITE`, `RECONFIGURED`, `UNRELATED`, and `UNSPECIFIED`.

Only tetrahedral, planar-bond, atrop-axis, and cumulene-axis shapes declare a binary opposite. Nonidentity square-planar, trigonal-bipyramidal, and octahedral relations are general reconfigurations. This is why a non-tetrahedral legacy `INVERTED` label can coexist with authoritative `reconfigured:<class>` evidence.

I Extended descriptors without a finite frame group

I.1 Cumulene axes

A cumulene-axis descriptor stores the full ordered axis path and two ordered two-reference terminal frames owned by the first and last path atoms. Validation requires at least three path atoms, pairwise distinct path atoms, integer path entries, exactly two terminal frames of width two, references distinct within each frame, and an *even* number of cumulated bonds. The even-bond requirement is the classical parity rule: an even run places the terminal substituent planes orthogonally and the element is axis-supported and chiral, whereas an odd run places them coplanar and the element is the bond-supported E/Z case.

I.2 Helical paths

A helical descriptor stores an ordered path of at least four atoms, an open/cyclic flag, a parity in $\{-1, +1\}$ or $\perp$, optional reported positions lying on the path, and an optional non-empty coupling identity. Identity is the canonical path form: for an open path, the minimum of the path and its reversal; for a cyclic path, the minimum over all rotations of the path and of its reversal. Equality compares descriptor class, canonical path, cyclicity, parity, and coupling identity.

Reported positions associate several reported atoms with one configuration and do not create independent local stereocentres; they are emitted in canonical path order. A fixed helical configuration may be constructed only through an authorized evidence source, and the declared-sidecar constructor requires an explicit `evidence_source` marker on deserialization, so bare connectivity cannot produce one. Reference replacement is refused on path atoms, which protects the locus in the same way that locus positions are protected in frame-based descriptors.

Table S6 Descriptor-class sets and the consumers that accept them.

Set	Members	Size
Legacy supported set (prefilter/rule consumers)	tetrahedral, square-planar, trigonal-bipyramidal, octahedral, planar-bond, atrop	6
Local configured (exact certificate)	supported set, plus cumulene-axis, extended cis/trans, helical, and planar chirality	10
Coupled framework (exact certificate/mirror)	fixed framework orientation with relative frame coupling; helper enumeration/focal operations refuse	1
Local enumeration and focal evidence	the ten local configured families; framework inputs deliberately refuse	10
Descriptor JSON restoration	all local configured families and framework	11
`RDKit` adapter	the supported set	6
Deferred (declared names only)	`rigid_bond_33`, `rigid_bond_23`, `rigid_bond_13`, `rigid_bond_12`	4

I.3 Descriptor-class sets

I.4 Exact auxiliary stereograph

For each configured registry, the exact encoder creates typed vertices for atoms, bonds, stereo loci, owner-specific ligand ports, virtual resources, allowed orbit tuples, and position-coloured tuple slots. Constitution and stereo are therefore canonicalized together by the native exact coloured-graph kernel. The certificate schema is `synkit.canonical-stereograph/2`. Equality compares complete canonical graphs; the associated SHA-256 digest is an index, not a substitute proof.

The pairwise classifier first compares constitution-only certificates. Equal constitutions are then compared as configured stereographs and, if unequal, after geometry-specific mirror transformation of the left registry. The outcomes are `IDENTICAL`, `ENANTIOMERS`, `DIASTEREOMERS`, `CONSTITUTIONALLY_DIFFERENT`, `INCOMPLETE`, and `UNSUPPORTED`, and `INFORMATION_MISMATCH`. RDKit drawing keys record their selected profile under `synkit.rdkit-stereograph/1`; chemical family keys use `synkit.resonance-family-stereograph/1`. Canonicalization and pairwise operations retain their Lewis-state defaults, while mirror assessment defaults to chemical

family identity. Select a common explicit profile when comparing these operations. The pairwise result schema is `synkit.stereoisomer-relation/2`.

J Proof-bearing computational entities

Table S7: Public molecular-layer value objects and evidence contracts.

Object	Contract
`StereoSupport` family	Immutable typed carrier with dependency set and relabelling that commutes with descriptor relabelling
`PotentialStereoElement`	Candidate carrier type, typed support, supplied-configuration state, provenance, and source identifier; no stereogenicity or configuration claim
`TetrahedralConstitution`	Center-fixed exact ligand orbits, multiplicities, automorphism count, and constitutionally-distinct or symmetry-related status
`StereoConfiguration`	Shape, specification level, concrete and canonical frames, relation and replacement operations
`HelicalStereo`	Path support, cyclicity, parity, reported positions, coupling identity, canonical path form
`PermutationWitness`	Concrete position permutation whose replay is checked
`StereoRelation`	Kind, double-coset class, and replayable witness
`GraphMorphism`	Immutable total injective material-node embedding with typed wildcard environment
`StereoMorphism`	Graph morphism plus local stereo certificates and explicit policy choices
`LocalStereoCertificate`	Layer, endpoint configurations, relation, witness, effective policy, and matched/ignored/propagate status

Object	Contract
`CIPRanking`	Ordered references, per-pair comparisons, completeness flag, and digest
`CIPAssignment`	Label or typed status, rankings, supplemental comparisons, projection witness, required rules, and digest
`MolecularChirality\-Result`	Classification, mirror isomorphism witness, descriptor counts, and evidence provenance

J.1 Stereo-preserving graph morphisms

`StereoMorphism` proves match compatibility; it does not classify a reaction-induced change. Material references are transported through the underlying injective map. Virtual hydrogen and lone-pair tokens retain their typed owner. A mapped explicit hydrogen removed by normalization is projected onto virtual H only when owner-local evidence makes the projection unique.

Every matched local certificate is checked twice: its witness must replay the transported frame, and its stored relation kind and class must agree with direct endpoint classification. Missing references, invalid virtual owners, non-injective maps, missing required targets, strict target extras, and forged witnesses are structured failures.

Table S8 Independent stereo-morphism policy axes.

Axis	Value	Meaning
Information	`exact`	Fixed matches equivalent fixed; unspecified matches unspecified
Information	`wildcard`	An unspecified query may specialize; a fixed query remains exact
Information	`either`	As wildcard; a fixed query may also use its declared compatibility alternative
Presence	`require`	Every source descriptor has one valid target; target extras allowed
Presence	`strict`	Require, and reject unmatched target descriptors touching the image
Presence	`ignore`	Accept structurally and emit explicit ignored certificates
Presence	`propagate`	Emit a visible obligation for a source descriptor without a target

Identity stereo morphisms contain identity node maps and witnesses. Composition pairs the concrete intermediate configurations, composes witnesses, replays the

composition on the transported source, and reclassifies the direct endpoint relation. The canonical signature includes shape, information level, status, policy, class, witness permutation, virtual-reference kinds, and wildcard-frame positions, and is invariant under consistent endpoint renumbering.

J.2 Value identity and hashing

For stereo-bearing graphs the map-invariant registry form built from connectivity/LLG colour-refinement tokens is a prefilter only. Equality enumerates exact structural isomorphisms and accepts one only if every registered descriptor aligns into its fixed group. Equal objects hash equally; equal hashes are never reported as isomorphism.

K Derived-label projection

K.1 Ranking kernel

Ligand ranking is computed from molecular constitution, isotopes, multiple-bond duplicate nodes, and ring closures. No toolkit CIP code or rank property is read. Per-ligand evidence records, for each sphere, the node's atomic number, mass, atom index, path, duplicate flag, duplicate source depth, and duplicate kind, together with one of three termination reasons: `exhausted`, `virtual_reference`, or `depth_cap`.

Rules are applied hierarchically: Rule 1a (atomic number, including averaged duplicate atomic numbers for simple non-fused even mancude rings) is exhausted over successive spheres before Rule 1b (duplicate-node distance), which precedes Rule 2 (isotopic mass). Rule 3 and the reference-independent, single-corresponding-pair subset of Rules 4c/5 are active. Each comparison records its outcome (`left_higher`, `right_higher`, `tie`, or `unsupported`), the first deciding rule, the deciding depth, and the compared witnesses on both sides. Multi-unit reference-descriptor pairing under Rule 4b and sequence-rule cases outside the implemented subset fail closed.

K.2 Assignment statuses

Table S9 Label-assignment statuses and their meaning.

Status	Meaning
`assigned`	A label is justified by a complete ranking and a witnessed projection
`unspecified_configuration`	The descriptor carries no orientation; no label may be emitted
`unresolved_priority`	The ranking returned a tie at the required depth
`unsupported_sequence_rule`	The comparison requires an unimplemented sequence-rule case, notably multi-unit Rule 4b
`unsupported_descriptor`	No CIP-style projection is claimed for this class (coordination geometries)

Axial M/P projection uses the IUPAC helicity convention from a witnessed viewing direction. When constitution cannot witness a stable viewing direction—a symmetric axis—the projection fails closed rather than using atom indices as a tiebreak. Assignments are immutable reports: they never mutate a molecule or descriptor, and their labels are absent from stereo identity, hashing, rule matching, and descriptor serialization.

L Whole-molecule chirality

Classification searches for a stereo-preserving isomorphism between the stereo-bearing graph and its descriptor-inverted mirror, refined by connectivity and LLG colours and by semantic feasibility on descriptors. A verdict of achiral returns the mirror map as its witness.

For underspecified input the outcome is four-valued: `necessarily_achiral`, `necessarily_chiral`, `configuration_dependent`, and `unsupported_or_incomplete`. Configuration dependence means at least one enumerated completion is chiral and at least one is achiral; it is a definitive property of the constitution. Enumeration is capped, mixed populations may terminate only after both classes have been witnessed, and caps without an independent proof fail closed to `unsupported_or_incomplete`. For an oversized all-tetrahedral assignment product, the implementation enumerates

the constitutional automorphisms instead. Transport into the descriptor-inverted mirror gives binary XOR constraints on the local configurations. A satisfying assignment is verified as an achiral witness; exhaustion of the automorphisms proves that no achiral completion exists.

Detection of a potential stereo locus from two-dimensional connectivity yields a typed `PotentialStereoLocus` carrying its axis support, an unspecified orientation state, its evidence provenance, and an unassessed stability status. Promotion to a fixed configuration requires an authorized evidence source: `explicit_stereo`, `validated_geometry`, or `declared_sidecar`. Molecular configuration evidence additionally carries a stability state (`unassessed` or `validated`) and an optional population fraction in $(0, 1]$, kept distinct from orientation.

Broad tetrahedral-carrier detection precedes stereogenicity. A carrier has four represented or hidden-hydrogen slots but may have repeated ligands. The constitution-only canonicalizer uniquely marks the center, enumerates exact attributed-graph automorphisms, and intersects their node orbits with the four ligand slots. Four singleton classes give `constitutionally_distinct`; any repeated class gives `symmetry_related`. The latter is deliberately not called achiral, because it means only that the configuration-neutral detector cannot confirm the center. Supplied configuration and CIP properties are not inputs and may not split or reorder the constitutional neighbor classes.

M Finite propositions and executable obligations

Let τ be a registered shape, $\Gamma_\tau \leq \Sigma_\tau$ its finite geometry group inside the frame symmetry group, and $[N]$ the orbit of a frame N.

P0—Configuration-neutral frame construction.

A local-frame constructor reads only constitutional attributes and is required to commute with attributed-graph relabelling. Exact support-stabilizer orbits are its tie

authority. Conjugation transports the stabilizer and its reference orbits under every graph isomorphism, proving that the tie certificate is independent of atom numbering, supplied configuration, and CIP. The executable tetrahedral slice currently validates the orbit partition; recursive total ordering of distinguishable classes is a subsequent implementation gate.

P1—Orbit canonicalization.

The minimum of the deterministically ordered orbit is unique, and two fixed configurations have the same canonical frame exactly when their frames share one orbit. Exhaustive atom-frame partitions contain 2, 3, 20, and 30 fixed orbits for tetrahedral, square-planar, trigonal-bipyramidal, and octahedral geometry; the bond and axis actions are also completely enumerated.

P2—Relabelling equivariance.

For bijective relabelling f, $[N] = [M]$ if and only if $[f(N)] = [f(M)]$, because position actions commute with value relabelling (Lemma S1).

P3—Replacement well-definedness.

An injective replacement of non-locus references that preserves frame distinctness descends to orbit classes. Missing sources, locus replacement, and reference collisions are rejected.

P4—Relation invariance.

The double coset $\Gamma_\tau \pi \Gamma_\tau$ is unchanged when either endpoint is replaced by an equivalent representative; relation kind and class ID are therefore representative independent. This is Lemma S8, and the gate checks it exhaustively for every registered action.

P5—Witness composition.

If p replays $A \to B$ and q replays $B \to C$, then `p.then(q)` replays $A \to C$; the global form of this law is the cocycle identity of Eq. (S2). Composition of stereo morphisms records and rechecks both legs and the direct endpoint claim; the companion paper extends the same obligation to reaction and fusion legs.

P6—Binary-opposite involution.

For tetrahedral, planar-bond, atrop-axis, and cumulene-axis shapes, $\mathrm{opp}(\mathrm{opp}([N])) = [N]$. These are exactly the index-two families of Corollary S7, which also shows why the three coordination geometries are not forced into a binary relation.

P7—Support commutation.

For every configured descriptor d and relabelling f, $\mathrm{support}(\mathrm{relabel}(d, f)) = \mathrm{relabel}(\mathrm{support}(d), f)$. Axis-support relabelling covers path atoms, material terminal references, and virtual-reference owners.

P8—Axis laws.

Every preserving terminal order and axis reversal yields an equivalent configuration; axis reversal and mirror action are involutions; an unspecified axis forgets terminal order but not its path; and reference replacement is refused on the full locus.

P9—Path laws.

Open-path reversal, and cyclic rotation together with reversal, are identity-preserving and involutive; mirror inversion changes only a fixed helical configuration; relabelling commutes with path support and canonicalization; reported positions are one coupled configuration rather than independent local centres; and coupling identity distinguishes two configurations on the same path.

P10—No inference from bare connectivity.

Two-dimensional topology detection produces only potential loci. Identical supports may carry opposite configurations only when an authorized evidence source declares them.

P11—Label independence.

Ranking is invariant under atom renumbering and fragment order; labels are invariant under renumbering, fragment order, valid support reversal, and descriptor canonicalization; mirror inversion changes the label exactly where the specification requires; toolkit CIP properties are never inputs; and no label is emitted from an unspecified configuration.

P12—Complete certificate invariance.

For every configured family, relabelling atoms and reconstructing the complete auxiliary stereograph preserves the exact certificate (Corollary 6); changing a fixed configuration changes it unless a whole-graph automorphism identifies the two states. CIP cache properties never enter vertex colours.

P13—Whole-stereograph relation partition.

After constitution equality is established, direct certificate equality, left-mirror equality, and residual inequality partition complete same- constitution pairs into identical, enantiomeric, and diastereomeric outcomes. Missing configuration and unsupported population semantics remain nondefinitive.

Table S10: Principal executable obligations.

Claim	Principal test
P1	`Test/Graph/Stereo/test_orbit_kernel.py::test_all_atom_frame_permutations_partition_into_expected_fixed_orbits` and the bond-frame orbit-size test
P2–P3	`test_relabeling_and_reference_replacement_are_equivariant` plus descriptor-migration locus protection
P4	`test_relation_class_is_independent_of_fixed_representatives`
P5	`test_witness_composition_replays_direct_transport`
P6	`test_binary_opposite_is_an_involution`
P7	`Test/Graph/Stereo/test_support_hierarchy.py::test_existing_descriptors_expose_typed_support_without_wire_changes` and `test_axis_support_relabeling_includes_path_material_and_virtual_owners`
P8	`Test/Graph/Stereo/test_axis_descriptors.py::test_every_preserving_terminal_order_and_axis_reversal_is_equivalent` and `test_axis_reversal_and_mirror_actions_are_involutions`
P9	`Test/Graph/Stereo/test_helical_descriptors.py::test_open_path_reversal_is_identity_preserving_and_involutive`, `test_cyclic_rotation_and_reversal_are_identity_preserving`, and `test_reported_positions_are_one_coupled_configuration_not_local_centers`
P10	`test_connectivity_detection_still_returns_only_potential_support` and `test_bare_connectivity_does_not_infer_helicity`
P11	`Test/Chem/Molecule/test_cip_ranking.py::test_atom_renumbering_preserves_the_rank_order`, `test_toolkit_cip_properties_are_not_inputs_to_ranking`, and `Test/Chem/Molecule/test_cip_assignment.py::test_atom_renumbering_and_fragment_order_preserve_the_label`

Claim	Principal test
P12	`Test/Graph/Stereo/test_configured_stereograph_families.py` and `test_canonical_stereograph_determinism.py`; public exhaustive relabelling report `exact_canonicalization_report.json`
P13	`Test/Graph/Stereo/test_pairwise_stereoisomer_relations.py` and `stereoisomer_relation_report.json`

N Validation corpus and conformance

Deterministic fixtures cover the registered shapes, the extended axis and path descriptors, support validation, symmetric isomorphisms, query policies, morphism policy axes, composition, CIP ranking and assignment, and serialization of descriptor payloads.

The configuration-free local totals comprise 6,999 theoretical local classes and 5,885 observed whole-graph classes. The difference of 1,114 is fully accounted for by exact global-symmetry quotients. Formal A, B, and C contain 48, 192, and 1,920 assignments, respectively. Raw A independently enumerates all 2,048 local encodings and produces the same 48 formal tuples and 38 global classes as formal A. The larger raw runs are stress evidence rather than additional assumptions of the correctness theorem. Raw B enumerates all 992,256 encodings and recovers the same 192 formal tuples and 95 classes as formal B. The raw C/VS146 stress run enumerates all 2,359,296 encodings for that record and recovers its 64 formal tuples and 64 classes. The other eight C raw products are deliberately absent from the maintained reproduction command; formal C nevertheless exhausts all nine cases by Corollary S13.

The optional `StereoMolGraph` bridge uses direct descriptor exchange. A legacy molecule-reconstruction diagnostic is retained only to show why bond-order recovery by another toolkit is not an appropriate acceptance criterion. `StereoMolGraph`

Table S11 Development-snapshot conformance results for the molecular-graph layer.

Gate	Result	Boundary
Full Pytest suite	3881 passed; 6 skipped; 179 passing subtests	Repository snapshot, 2026-07-26
Orbit, support, axis, path, and plane laws	Exhaustive enumeration for all registered actions; P7–P12 fixtures pass	Algebraic method boundaries
All-family local quotient	940 raw representations yield 67/67 expected classes	Ten configured families
All-family exact certificates	67/67 classes distinct within family; 0 collision or timeout	Every configured class
Configuration-free local corpus	66,468/66,468 representations; 3,476/3,476 carriers	1,218 Internal, ACS, CIP, and RotA rows
Formal mixed-family A/B/C	2,160/2,160 assignments; 30/30 cases; 38/95/752 classes	Simultaneous carrier configurations; no failure or timeout
Raw mixed-family A/B/C slice	3,353,600/3,353,600 encodings over 22 scoped cases	Complete A and B plus complete VS146 slice of C; no failure or timeout
Synthetic whole-graph relabelling	201 per-class plus 528 representative checks; 0 failures	Exhaustive 5! for four small fixtures, sampled otherwise
Mirror closure	67/67 classes closed; 7 fixed	Three square-planar, two planar-bond, plus two extended-cis/trans classes
Public ACS certificate audit	888/888 atom permutations invariant; 4/4 CIP-cache removals invariant	Four small configured records
Multi-element composition	32 assignments yield 31 classes; 13 enantiomer and 64 diastereomer pairs	Six designed graphs; every class mirrored
Multi-element ACS audit	14/14 sampled renumberings; 7/7 mirror labels	Seven records with 18 configured elements
Pairwise relation matrix	9/9 correct	P13 designed conformance cases

remains absent from runtime dependencies. The pre-split 300/300 bridge and 6/6 conformance totals are not current evidence and must be regenerated before inclusion in the submission artifact.

O Frozen chemical annotation and replay

The source-selection manifests record retrieval, terms, exact checksums, constitution keys and annotation conventions. The coordination and negative selections are outside the ACS/CIP/RotA development constitutions. The RotA sample is explicitly

drawn from that development data. Original annotations remain archived byte-unchanged. The clean-release replay regenerates annotations and production reports, binds their SHA-256 checksums, and reproduces every scientific summary field. The manuscript summary recomputes its counts from those four fresh reports; their execution provenance is separately recorded. Selection, annotation, production evaluation and mathematical reference coverage therefore remain separate audit steps.

For coordination, the first source conformer is used without selecting the best later conformer. Donor-slot isotope labels are removed before assigning the physical constitutional profile, while the supplied local tetrahedral annotations are transported into the reference and production inputs. The independent octahedral fit minimizes normalized RMSD over proper rotations and all donor-slot assignments. Its acceptance limits were fixed before evaluation: normalized RMSD at most 0.35 and a gap of at least 0.02 to the next inequivalent pose. The same criteria apply to the reflected coordinates. The two refusals are `Mn_PNhP_iPr_tri_0` and `Ru_NNhN_iPr_tri_0`, with RMSDs approximately 0.490 and 0.489. They remain in the selected denominator and are not recast as incorrect production predictions. The 38 certified examples retain 18 local frames; their independent source/mirror relations are 29 achiral and 9 chiral. These are configured-graph relations under the declared retained information.

Direct reference enumeration covers all 900 formal assignments. Production replays every reference partition and mirror map, then compares the certificates of certified source, source-renumbered, coordinate-mirrored and production-mirrored states. The resulting equality relation must neither split one reference class nor merge different classes. The 47 observed classes are therefore a subset of the 230 reference classes, rather than a claimed evaluation of every generated reference pose.

The sourced biaryl negatives are retained with original retrieval responses and failed-query exclusions. Their truth witnesses preserve the complete typed graph, fix the two axis endpoints, and exchange a declared terminal ligand pair. The same map

is replayed before original/renumbered perception. This checks a constitutional counterexample to axis confirmation; it neither estimates a kinetic barrier nor labels a compound experimentally achiral.

The RotA annotation contains every frozen sample locus and records uncertainty without replacing difficult cases. Of 52 axis supports, 26 admit the terminal-exchange witness and 26 exhaust the bounded search without one. This terminal criterion differs from global configured identity: a simultaneous exchange at both ends can preserve a coupled axis orientation. The complete per-case assignments, partitions, mirror involutions and transport composition checks define the reference used in the production comparison. For atom pairs and spiro categories, geometry does not by itself confirm the focal tetrahedral support. Negative tetrahedral perception may omit a symmetry-related center, and the category projection requires the actual focal carrier status. All 120 original/renumbered perception comparisons are checked against those frozen expectations. The 98 reported classes are summed over cases and are not a count of distinct chemical compounds. Experimental persistence remains unknown throughout this sample.

P Local perception and naming benchmarks

All 1208 records are invariant under reverse atom renumbering independently in both settings. The erased setting misses 264 ACS and 283 CIP positions; all 547 are carbon tetrahedral carriers classified as symmetry-related by the configuration-erased center stabilizer and all are recovered in the retained setting. ACS contains 134 acyclic and 130 ring misses; CIP contains 137 acyclic and 146 ring misses. Available neighbor evidence comprises 510 tetrahedral-only, 14 planar-only, 22 mixed tetrahedral–planar, and 1 cumulene case.

Additional candidates are serialized per record. Their retained/erased counts are 109/52 for ACS and 57/19 for CIP. RotA has 1562 additional broad typed projections

Table S12 Task-specific two-setting local validation. "Retained" uses supplied orientation only at other local-neighbor frames; "erased" removes all configuration. CIP positions and labels are not global chirality truth; RotA is positive-only.

Task	Retained	Erased	Reference	Additional evidence
ACS configured source loci	1007	743	1007	109/52 additional candidates
CIP all local positions	1252	969	1252	57/19 additional candidates
tetrahedral	980	697	980	Exact center-stabilizer criterion
planar	232	232	232	Typed planar carriers
axial/helical	40	40	40	No stability claim
RotA annotated typed loci	698	698	698	1562 broad additional projections
RotA expanded cumulene paths	15	15	15	Modulo path reversal
Reverse-renumbered records	1208	1208	1208	Complete invariance in each setting
CIP coordinate-assisted exact records	187		300	926/1252 labels; 937 emitted

in either setting. Restricting to constitutionally confirmed carriers recovers 688/698 annotated loci and leaves 833 additional projections. Confirmed spiro-category projections require actual confirmed tetrahedral supports; geometry alone does not supply confirmation. These values are annotation-list differences, not false-positive rates: ACS local annotations are incomplete and RotA is positive-only. They make no handedness, configurational-stability, or whole-molecule chirality claim.

A separate synthetic negative protocol contains 24 constitutional controls. 13 axis-like cases have exact axis-fixed automorphisms exchanging a terminal ligand pair, and 11 have no supported axis topology. Every case is invariant under reverse atom renumbering and the confirmed-axis false-positive count is zero. Because these are constitutional proofs rather than kinetic measurements, they do not validate rotational barriers, isolation timescales, stability, or handedness.

CIP source versions.

The provided publisher supporting archive (DOI https://doi.org/10.1021/acs.jcim.8b00324.s001) is unchanged and is licensed under CC BY-NC 4.0. Preparation repairs citation punctuation in VS297 and VS298 only; all 300 molecular inputs and expected

labels are unchanged, and the resulting SMILES file matches the evaluated checksum. The original publisher SDF differs from the later GitHub SDF at VS095 (atom ordering and conformer) and in those two citation fields. Original and amended coordinates have separate accepted checksums. A fresh full 300-record evaluation of the original SDF reproduces every native naming record and every RDKit comparator record from the amended-SDF evaluation; the primary numerical manifest now identifies that original-source report. This agreement does not assert coordinate equivalence. The publisher grant is asserted only for the provided original data and documented citation repair.

The pinned-coordinate CIP naming protocol has 73.96% micro-label recall and 98.83% precision; the SMILES-only exact-record baseline remains 175/300. The 113 non-exact records are retained as 83 ranking defects, 21 missing- orientation cases, and 9 unsupported classes. Rules 1a/1b/2, simple mancude-ring averaging, Rule 3, and the reference-independent one-pair subset of Rules 4c/5 are active; multi-unit Rule 4b remains fail-closed. The input-contract audit also identifies a strict information boundary: VS010 and VS011 have the same supplied SMILES but opposite helical labels, so SMILES-only exact-record accuracy is bounded by 299/300. In the pinned 3D file, atom positions 1–4 have opposite nonzero signed normalized volumes for the two records. This translation- and proper-rotation-invariant, reflection-sensitive witness makes the pair distinguishable without reading its reference labels. The coordinate protocol completes both helical records, all five CT4 records, four of seven AT records, and the fully material-framed VS144 cumulene; the remaining sequence rules and extended classes remain incomplete.

Table S13 Supplied-stereo protocol on the 258-record ACS validation set. Comparator columns are the publisher's recorded results. "Agreement" is cross-profile agreement with the ACS topology labels, not internal certificate correctness. ACS labels: 94 achiral, 164 chiral.

Procedure	Agreement	Fraction
Topology completion (this work)	258/258	1.000
Exact + necessary global certificate	257/258	0.996
`StereoMolGraph` (published)	258/258	1.000
InChI (published)	238/258	0.922
`RDKit` SMILES (published)	235/258	0.911
chython (published)	210/258	0.814
Exact supplied stereograph	253/258	0.981

Q Whole-molecule chirality benchmark

The exact-stereograph row is a separate 5 s per-case supplied-configuration audit. It preserves every explicit source `@`/`@@` atom configuration and leaves additional undeclared potential loci unconstrained. Chemical identity compares the exact resonance family while retaining genuine bond-order differences. It gives 255 definitive outcomes, of which 253 agree, and times out on VS126, VS127 and VS208. VS170 is chiral under chemical identity because its `S-I` and `S#I` ligands remain distinct, whereas the ACS topology profile omits bond order. In VS300 the exact certificate mirror fixes the central configured atom and exchanges the two configured cages; the ACS result instead depends on global topology-completion probes outside the configured support catalogue. This audit makes no configured-locus completeness claim and must not be presented as the source of the topology-completion 258/258 result.

The separately versioned exact-plus-global protocol uses a 30-second budget for source-declared identity and adds a map-independent, orientation-unspecified coupled-frame certificate. It proves VS300 necessarily chiral and therefore agrees with 257/258 ACS labels. It does not choose a VS300 enantiomer and does not alter the frozen source-declared report; VS170 remains a declared chemical identity/profile difference.

Coverage asks whether the erased manual label remains among the possible configurations; it is not recovery accuracy, because the input configuration was deliberately destroyed. VS226 exhausts after six symmetry-unique isomers and is necessarily

Table S14 Stereo-stripped four-state protocol on the same set, 256-isomer cap. The 258 records reduce to 214 unique constitutions.

Outcome	Rows
Necessarily chiral	125
Necessarily achiral	66
Configuration dependent	67
Unsupported or incomplete	0
Definitive rows	258
Erased manual label still in the possible population	258

achiral. Exact automorphism-parity analysis proves VS265 and VS266 necessarily chiral and supplies both class witnesses for configuration-dependent VS268, without enumerating their multi-million-entry raw assignment spaces.

The RotA and CIP datasets are not scored as chirality benchmarks. Historical binary columns for those datasets used an exploratory arbitrary-axis orientation and carry a machine-readable retraction; they must not be cited as current classification or accuracy.

R Reproduction commands

```
# from repository root in the locked Linux/Python 3.12 environment
PYTHON=python bash scripts/lint.sh
PYTHON=python bash scripts/pytest.sh -q Test/
python -m Experiment.Stereo.reproduce --tier bounded --jobs 2 \
  --output-dir /tmp/stereo-bounded
python -m Experiment.Stereo.reproduce --tier full --jobs 2 \
  --stereomolgraph /path/to/pinned/StereoMolGraph \
  --output-dir /tmp/stereo-full


# compile this paper
cd paper/stereo_graph
latexmk -pdf main.tex
```

The release instructions specify the wheel-hash environment lock, the separate optional-comparator dependency overlay and exact comparator commit. The complete 35-stage clean execution retains every report and stage log. Optional raw Cartesian stress is excluded from this sequence and remains separately identified historical supplemental evidence.

S Known limitations

- The local descriptors do not infer coordinates, conformations, configurational stability, reaction selectivity, or preferred paths.
- Planar chirality and coupled tetrahedral framework orientation are configured. Additional global geometry families and configured orientation perception remain unsupported. The analyzer fails closed above 256 support atoms or 64 coupled frames and for disconnected frame-bearing components. The four `rigid_bond` variants are class names only, and enhanced stereo groups remain deferred.
- The exact certificate consumes all ten configured classes. Cumulene-axis, helical, and planar-chiral descriptors remain outside the narrower legacy graph-identity, rule, JSON/GML, and `RDKit` surfaces.
- Exact configured canonicalization currently materializes complete local orbits and uses verified automorphism pruning. The retained ACS audit has no 2 s timeout, but larger raw local products remain outside the maintained benchmark.
- The RotA corpus is positive-only. Full annotated-locus recovery therefore does not validate the 1,562 additional broad axis candidates, their configurational stability, or their handedness. The 24 synthetic negatives validate constitutional specificity but not those experimental properties.
- RDKit round-trips the six supported classes and assigned non-tetrahedral/atrop states; unsupported unknown orientations fail explicitly.

- The CIP kernel implements Rules 1a, 1b, and 2, simple mancude-ring averaging, Rule 3, and the reference-independent one-pair subset of Rules 4c/5. Multi-unit Rule 4b and coordination-geometry projections remain fail-closed. Full 300-record evaluation additionally requires coordinates or declared orientation evidence because two supplied SMILES are identical but configurationally opposite.
- Local finite-group evidence does not prove equivalence of `SynKit`'s operational graph matching and a full DPO system.

T Attribution and reproducibility checklist

The ordered local-list and group-action formulation follows Andersen, Flamm, Merkle, and Stadler [1]. The six-class relative descriptor layer independently implements ideas from Papusha and Leonhard [2]. Non-tetrahedral tables were cross-validated against the MIT-licensed `StereoMolGraph` commit named in Table S3; the retained notice is `LICENSES/StereoMolGraph-MIT.txt`. The CIP sequence-rule hierarchy follows IUPAC recommendations P-92. The ACS validation set is redistributed under CC BY-NC 4.0 and the RotA corpus under the MIT License. The original CIP publisher inputs are included under CC BY-NC 4.0 with attribution and two explicitly documented bibliographic punctuation repairs.

The research archive retains the exact executed source and input checksums, the environment lock, independent fixtures, unchanged raw execution reports and the numerical manifest used to generate manuscript values. Package version, source commit and report schema identify distinct parts of this provenance.

References

[1] Andersen, J.L., Flamm, C., Merkle, D., Stadler, P.F.: Chemical graph transformation with stereo-information. In: Graph Transformation (ICGT 2017). Lecture

Notes in Computer Science, vol. 10373, pp. 54–69. Springer, ??? (2017). https://doi.org/10.1007/978-3-319-61470-0_4

[2] Papusha, M., Leonhard, K.: StereoMolGraph: Stereochemistry-aware molecular and reaction graphs. Journal of Chemical Information and Modeling **66**(7), 3830–3839 (2026) https://doi.org/10.1021/acs.jcim.5c02523 . Optional SynKit implementation comparison pinned at commit 2189f610f23eaaf992e2e01a12ea4d0532496601

[3] Papusha, M., Leonhard, K.: A Stereochemistry-Aware Circular Algorithm and Fingerprint for Resonance-Invariant Molecular Representation. ChemRxiv preprint, version 2. Posted 21 July 2026; comparison limited to accessible primary abstract and author experiment source (2026). https://doi.org/10.26434/chemrxiv.15002723/v2

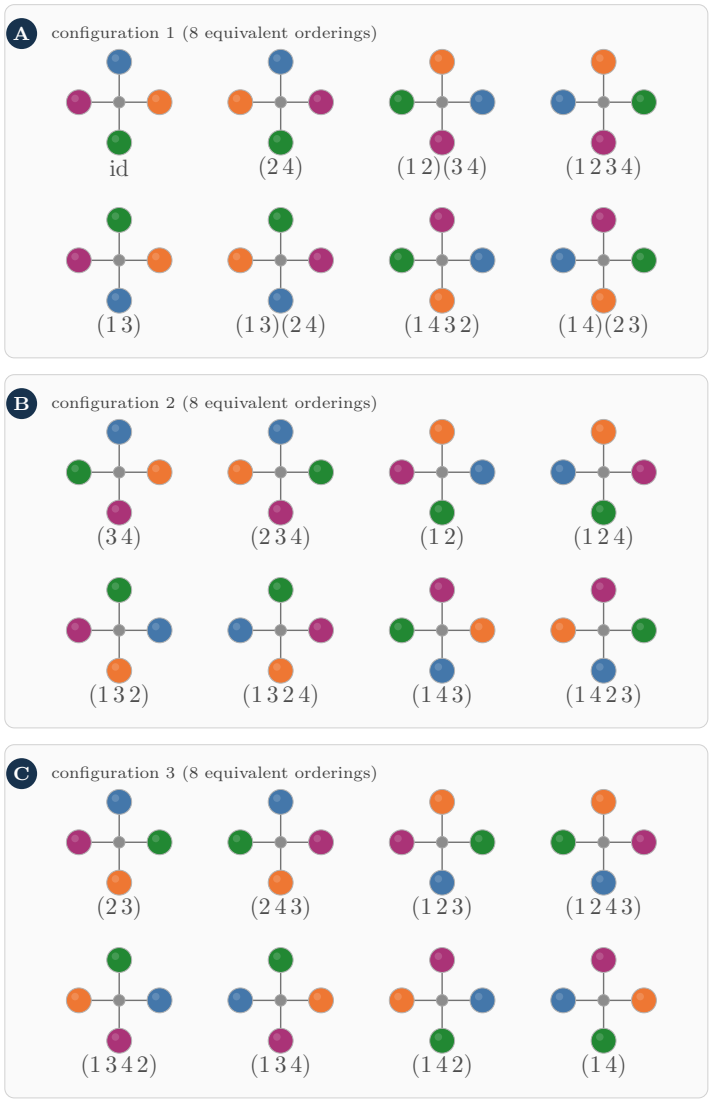


Fig. S5 Square planar. All 24 orderings of four distinct ligands, grouped by configuration. Here $\Sigma_\tau = S_4$ and $\Gamma_\tau = D_4$ of order 8, so each configuration is an orbit of 8 orderings and there are $[\Sigma_\tau : \Gamma_\tau] = 24/8 = 3$ configurations; unlike the tetrahedral case there is no single binary "opposite", which is why nonidentity square-planar relations are general `RECONFIGURED` classes.

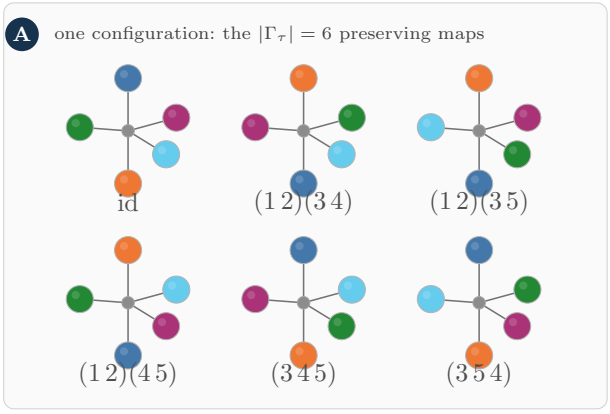


Fig. S6 Trigonal bipyramidal. The six orderings forming one configuration (the $\Gamma_\tau = D_3$ orbit: an axial pair and an equatorial triple). With five distinct ligands there are $[\Sigma_\tau : \Gamma_\tau] = 120/6 = 20$ configurations in total, so only one 6-element orbit is drawn.

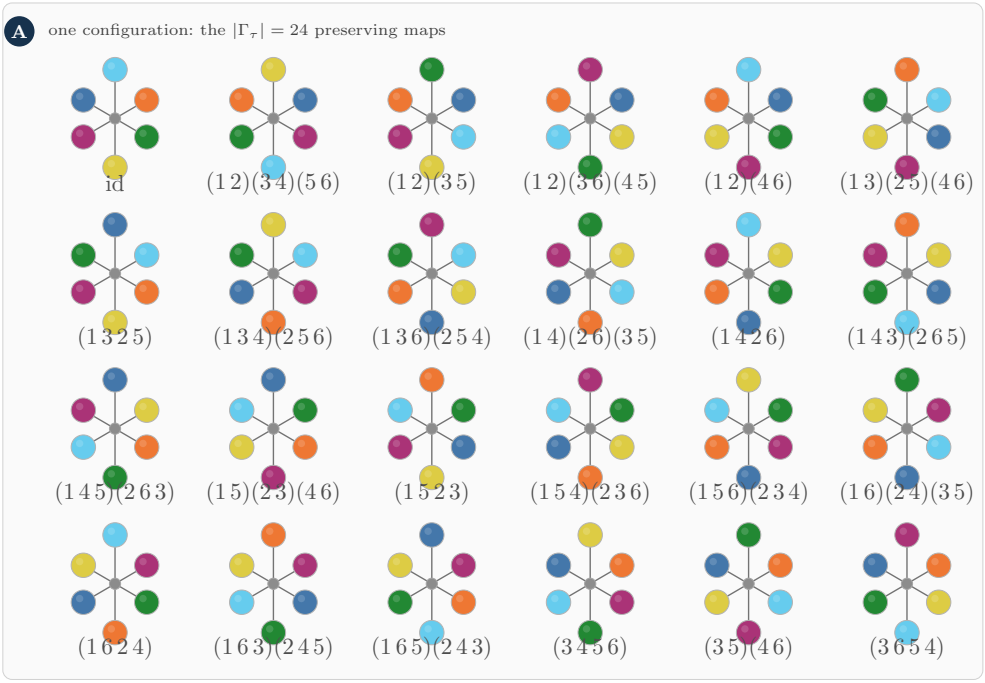


Fig. S7 Octahedral. The 24 orderings forming one configuration (the proper rotation group $\Gamma_\tau = O$). With six distinct ligands there are $[\Sigma_\tau : \Gamma_\tau] = 720/24 = 30$ configurations in total (the classical $[Mabcdef]$ stereoisomer count); only one 24-element orbit is drawn.

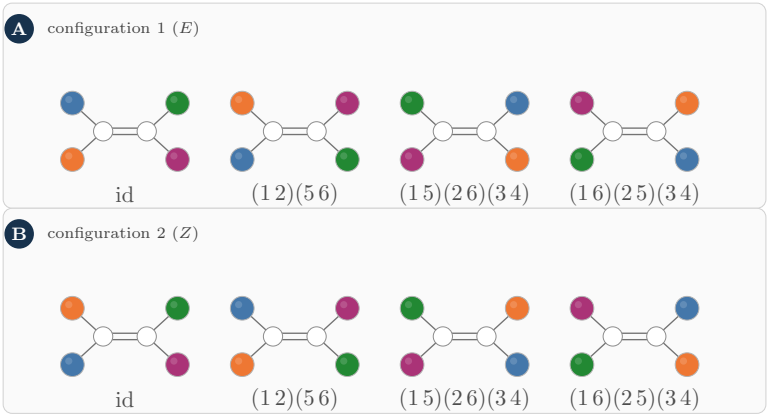


Fig. S8 Planar bond (E/Z). The eight bond-frame maps split into two orbits of four; each orbit is one configuration. Here $\Gamma_\tau = V_4$ of order 4 sits inside the bond-frame group Σ_τ of order 8, so $[\Sigma_\tau : \Gamma_\tau] = 2$ and a single admissible one-end swap realizes the binary $E \leftrightarrow Z$ opposite of Corollary S7.

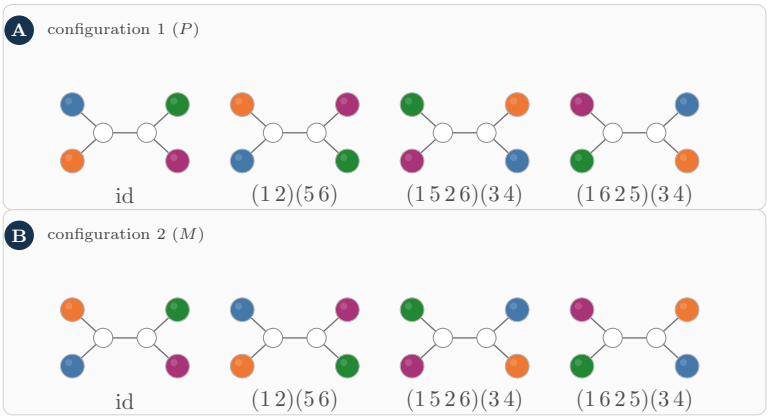


Fig. S9 Atrop axis. The eight frame maps split into two orbits of four; each orbit is one assigned axial sense. Again $\Gamma_\tau = V_4 \leq \Sigma_\tau$ with $[\Sigma_\tau : \Gamma_\tau] = 2$, realizing the binary $P \leftrightarrow M$ opposite; unlike the planar bond, the improper extension here also has index two, which is why the axis is chiral. The frame is drawn with a single bond to distinguish it from the planar-bond (π-bearing) frame of Figure S8. The cumulene-axis frame shares this structure over an even-cumulene path.